

Estimation and inference in discrete games with uncertain behavior

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Abstract

We revisit the econometric analysis of normal-form discrete games where, instead of assuming that all observations come from the same behavioral model (e.g, Nash equilibrium), we pre-specify a collection of *candidate* behavioral models (each one with potential multiple solutions) that could have generated the data, with different observations possibly having been produced by different behavioral models. Assuming the existence of an observable *instrument* Z that can capture the dependence between the behavior/solution selection mechanisms and payoff covariates, our global model becomes a semiparametric convolution of the candidate behavioral models. We show conditions under which the parameters of the model are point-identified, and we propose conditional-GMM estimation procedure and we evaluate its performance through Monte Carlo experiments. As an empirical illustration, we analyze geographic entry decisions by Lowe's and Home Depot in the continental United States. Assuming coalitional cooperation and Nash equilibrium as the two candidate behavioral models, we find evidence consistent with the presence of both types of behavior across markets, with the probability of cooperation decreasing with market size.

Keywords: Estimation of games, uncertain behavior, multiple equilibria, semiparametric convolution.

JEL classification: C1, C14, C57.

1 Introduction

One of the main goals of the econometric analysis of games is the estimation of players' reduced-form payoff functions. This requires assuming a behavioral theory (e.g, Nash equilibrium) that links the outcomes observed to the underlying game. Thus, the behavioral model assumed is key to the validity of the analysis, and having inferential methods

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that provide some degree of behavioral robustness is a goal we should pursue. Allowing for behavioral heterogeneity can be facilitated in experimental settings where the researcher has precise knowledge about payoff functions (see, e.g., Crawford, Costa-Gomes, and Iriberri (2013)), but it is typically more challenging with nonexperimental data where payoff functions themselves are unknown and must be estimated. As a result, the econometric analysis of games in nonexperimental settings typically relies on the assumption that every observation in the data was produced by the same behavioral model, which is assumed to be known (e.g., Nash equilibrium). Focusing on static discrete games, we propose an econometric methodology that allows for behavioral ambiguity. Instead of assuming one behavioral model, we pre-specify a collection of *candidate* behavioral models, each one with potentially multiple solutions, that could have produced each observation in the data. Latent behavior and solution-selection mechanisms determine which behavioral model and which corresponding solution are selected for each observation.

Assuming the existence of an observable *instrument* Z that captures the statistical dependence between the selection mechanisms and payoff covariates, the predictions of our global model can be written as a convolution of the predictions of the candidate solutions, with the convolution weights being nonparametric functionals of Z . If our candidate solutions are not observationally equivalent, the weights of the convolution can be semiparametrically identified, and point-identification of the payoff parameters can follow if the latent selection mechanisms assign nonzero probability to behavioral models that have sufficient identification power. In the absence of such restrictions, a CS for the parameters can be constructed from our semiparametric convolution. Since our semiparametric convolution implies conditional moment restrictions, we propose a conditional-GMM estimator. Our setup and method allows us to recover the normal-form payoff parameters as well as the convolution weights, which contain information about the propensity of the selection mechanisms to select each behavioral model, and each solution within each behavioral model. Our framework relies on the availability of an instrument Z that satisfies an exclusion restriction, but this assumption (and our model) can be tested through a variety of existing specification test approaches.

The paper proceeds as follows. Section 2 introduces our approach through a 2×2 game with two candidate behavioral models: complete-information pure-strategy Nash equilibrium and (coalitional) cooperation. Section 3 expands our model by allowing for mixed strategies. Section 4 goes beyond the 2×2 game and describes a general discrete game with $P \geq 2$ players. In each section we discuss identification and estimation of the normal-form parameters and of the behavioral weights, which reveal the propensity of the latent selection mechanisms to select each behavioral model and each solution within each behavioral model. In each case we propose conditional-GMM estimators based on

the semiparametric convolution of our global model. The properties of our approach are studied in Section 5 through Monte Carlo experiments. Section 6 includes an empirical illustration where we apply our method to the analysis of geographic-market entry decisions by Lowe’s and Home Depot. In this exercise we use coalitional cooperation and pure-strategy Nash equilibrium as the candidate behavioral models, and we find evidence consistent with the presence of both behavioral models across different markets, with the probability of cooperation decreasing with market size. Section 7 concludes. Appendix A includes details of our main econometric results, as well as results of additional Monte Carlo experiments. An Econometric Supplement¹ includes step-by-step derivations of our econometric results.

2 A 2×2 game

Consider the following simultaneous 2×2 normal-form game,

Table 1S-. A game of strategic substitutes

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	$t_1 - \Delta_1 - \varepsilon_1, t_2 - \Delta_2 - \varepsilon_2$	$t_1 - \varepsilon_1, 0$
$Y_1 = 0$	$0, t_2 - \varepsilon_2$	$0, 0$

Consider the case of strategic substitutes ($\Delta_p \geq 0$ for $p = 1, 2$) and complete-information, where both players observe $(t_p, \Delta_p, \varepsilon_p)_{p=1}^2$ before making their choices. We will allow for mixed strategies and more general discrete games in subsequent sections. Let $\mathcal{Y} \equiv \{0, 1\} \times \{0, 1\}$ denote the action space, with $y \equiv (y_1, y_2) \in \mathcal{Y}$ denoting an element of $\mathcal{Y}$, and $Y \equiv (Y_1, Y_2)$ denoting the actual outcome of the game. Suppose we consider two candidate behavioral models,

- (i) **Noncooperative:** Players play a pure-strategy Nash equilibrium (PSNE).
- (i) **Cooperative:** Players engage in cooperative (coalitional) bargaining and select the outcome $Y \equiv (Y_1, Y_2)$ that maximizes the sum of their payoffs (i.e, the total value of the coalition).

With transferable utility, maximization of joint payoffs is the prediction of Nash bargaining (see Myerson (1990), Thomson (1994), or Watson (2013, Appendix D)). A solution to each model is an outcome Y consistent with its behavioral predictions. Figure 1 fixes $(t_p, \Delta_p)_{p=1}^2$ and shows the solutions of each behavioral model across different realizations of $(\varepsilon_1, \varepsilon_2)$ in $\mathbb{R}^2$. Cooperation always yields a unique prediction for Y , while PSNE produces multiple predictions in some regions, which is a well-known result (see Bjorn and Vuong

¹https://aaradill.github.io/econometric_supplement_uncertain_behavior.pdf

(1984), Bresnahan and Reiss (1990), Bresnahan and Reiss (1991b), Bresnahan and Reiss (1991a), Tamer (2003), Berry and Tamer (2007)). However, as shown in Figure 2, both models produce a unique prediction for $S \equiv Y_1 + Y_2$ for any realization of payoff shifters. For this reason, we will focus the analysis in this section on the aggregate outcome S . This will allow us to bypass the properties of the PSNE equilibrium selection mechanism in the region of multiple equilibria, and focus the analysis on the properties of the behavioral selection mechanism that chooses between PSNE and cooperation. Subsequent sections will model both behavioral and solution selection mechanisms in more general games.

2.1 A parameterization of normal-form payoff functions

We will assume that $\varepsilon \equiv (\varepsilon_1, \varepsilon_2)$ are unobserved by the econometrician, and we will model the remaining payoff shifters $(t_p, \Delta_p)_{p=1}^2$ as follows. Let $(X_1^{ns}, X_2^{ns}, X_1^s, X_2^s)$ be a collection of covariates that are observable to the econometrician. X_p^{ns} includes covariates that shift² player p 's non-strategic payoff component t_p , while X_p^s denotes covariates that shift player p 's strategic-interaction payoff component Δ_p . Let $(\beta_{10}, \beta_{20}, \Delta_{10}, \Delta_{20})$ be unknown parameters. We will parameterize $t_p = X_p^{ns'} \beta_{p0}$ and $\Delta_p \equiv X_p^{s'} \Delta_{p0}$ for $p = 1, 2$, so the normal-form representation of the game is given by,

Table 2S-. A parameterized game of strategic substitutes

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	$X_1^{ns'} \beta_{10} - X_1^{s'} \Delta_{10} - \varepsilon_1, X_2^{ns'} \beta_{20} - X_2^{s'} \Delta_{20} - \varepsilon_2$	$X_1^{ns'} \beta_{10} - \varepsilon_1, 0$
$Y_1 = 0$	$0, X_2^{ns'} \beta_{20} - \varepsilon_2$	$0, 0$

The values of $(\Delta_{10}, \Delta_{20})$, and the support of (X_1^s, X_2^s) are such that $X_p^{s'} \Delta_{p0} \geq 0$ w.p.1, so we have a strategic-substitutes game. Constant strategic interaction effects corresponds to $X_p^s \equiv 1$ and $X_p^{s'} \Delta_{p0} = \Delta_{p0}$. Nonstrategic X_p^{ns} and strategic X_p^s payoff shifters can have elements in common for each player and across players, as long as the conditions we will describe below are satisfied. We will group $X \equiv (X_1^{ns} \cup X_2^{ns} \cup X_1^s \cup X_2^s)$ and maintain that players have complete information, so they observe the realization of (X, ε) , and they know the true values of the payoff parameters.

2.2 A parameterization for the joint distribution of ε

This paper will focus on games where both the normal-form payoff functions as well as the distribution of unobserved payoff shifters are parametrically specified. In our 2×2 game, this will require a parametric specification for the distribution $\varepsilon|X$. To this end, we introduce the following assumption.

²We will use the terms “payoff covariates” and “payoff shifters” interchangeably.

Figure 1: Predicted outcomes for Y under cooperation and PSNE

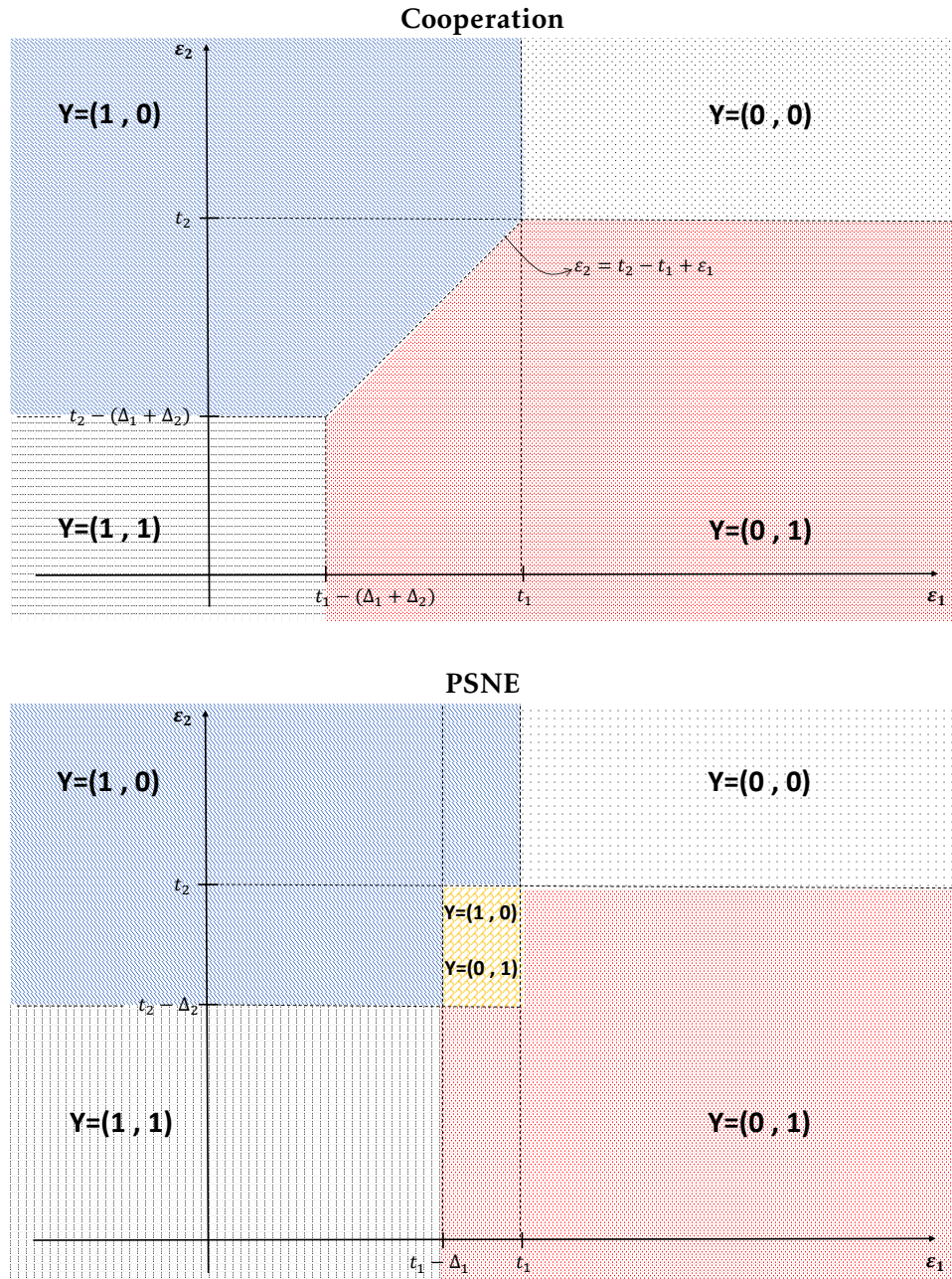
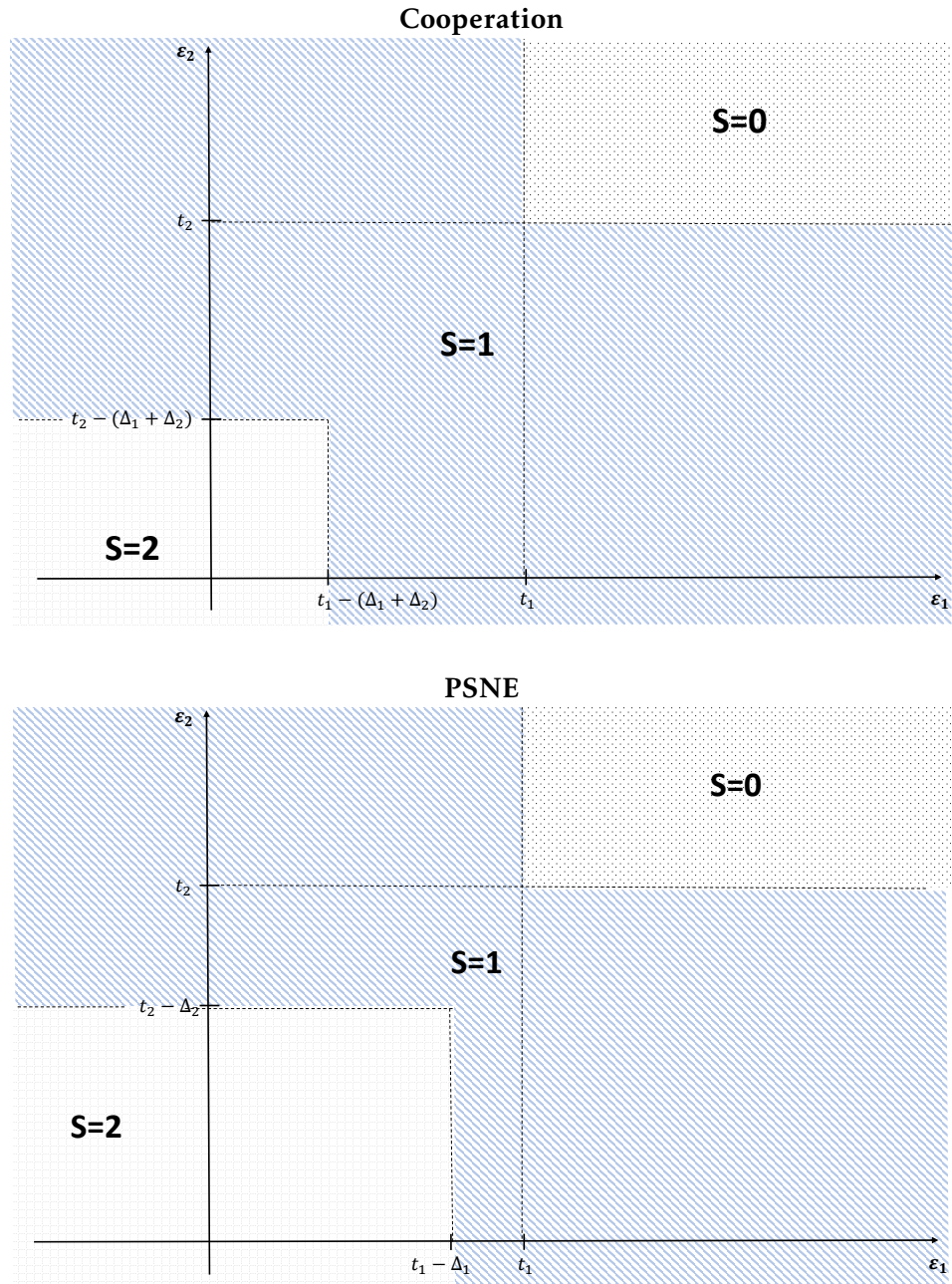


Figure 2: Predicted outcomes for $S \equiv Y_1 + Y_2$ under cooperation and PSNE



Assumption G1 (A parametric distribution for ε)

(i) $\varepsilon \sim F_{1,2}(\cdot, \cdot | \rho_0)$, a parametric, jointly continuous distribution with unbounded support on $\mathbb{R}^2$ indexed by a scalar parameter ρ_0 that captures the interdependence between ε_1 and ε_2 . We assume that ε are independent of X , although this can be relaxed if we parameterize the conditional distribution of $\varepsilon|X$. We denote the marginal distribution of ε_p as $F_p(\cdot)$.

(ii) Group the non-strategic parameters as $\gamma_0 \equiv (\beta_0, \rho_0)$, with $\beta_0 \equiv (\beta_{10}, \beta_{20})$, and the strategic-interaction parameters as $\Delta_0 \equiv (\Delta_{10}, \Delta_{20})$. We will denote $\theta_0 \equiv (\gamma_0, \Delta_0) \in \Theta$, where the parameter space Θ is assumed to be compact. We will denote a generic element in Θ as $\theta \equiv (\gamma, \Delta)$, with $\gamma \equiv (\beta_1, \beta_2, \rho)$ and $\Delta \equiv (\Delta_1, \Delta_2)$.

(iii) Let $\nabla_p F_{1,2}(\varepsilon_1, \varepsilon_2 | \rho) \equiv \frac{\partial F_{1,2}(\varepsilon_1, \varepsilon_2 | \rho)}{\partial \varepsilon_p}$. Then, $|\nabla_p F_{1,2}(\varepsilon_1, \varepsilon_2 | \rho)| \leq \overline{D} < \infty \forall (\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2, \rho \in \Theta$.

■

Since both of our candidate behavioral models are characterized by threshold-crossing decision rules, the scale of ε_p cannot be separately identified from the scale of the payoff parameters. For this reason, the variance of ε_p is fixed in $F_p(\cdot)$. In our strategic-substitutes game, we will restrict Θ to satisfy $X_1^{s'} \Delta_1 \geq 0$ and $X_2^{s'} \Delta_2 \geq 0$ w.p.1 $\forall \Delta \in \Theta$.

2.3 An expression for $E[S|X]$ in each behavioral model

For a given (θ, X) , the regions of $\varepsilon \in \mathbb{R}^2$ that predict each possible value of $S \in \{0, 1, 2\}$ for our candidate behavioral models are (see Figure 2),

Cooperative-behavior regions:

$$\begin{aligned} \mathcal{R}_S^C(0|X, \beta) &\equiv \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : \varepsilon_1 \geq X_1^{ns'} \beta_1, \varepsilon_2 \geq X_2^{ns'} \beta_2\}, \\ \mathcal{R}_S^C(2|X, \beta, \Delta) &\equiv \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : \varepsilon_1 \leq X_1^{ns'} \beta_1 - (X_1^{s'} \Delta_1 + X_2^{s'} \Delta_2), \varepsilon_2 \leq X_2^{ns'} \beta_2 - (X_1^{s'} \Delta_1 + X_2^{s'} \Delta_2)\}, \\ \mathcal{R}_S^C(1|X, \beta, \Delta) &\equiv \mathbb{R}^2 \setminus (\mathcal{R}_S^C(0|X, \beta) \cup \mathcal{R}_S^C(2|X, \beta, \Delta)). \end{aligned}$$

Noncooperative-behavior regions:

$$\begin{aligned} \mathcal{R}_S^{NC}(0|X, \beta) &\equiv \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : \varepsilon_1 \geq X_1^{ns'} \beta_1, \varepsilon_2 \geq X_2^{ns'} \beta_2\}, \\ \mathcal{R}_S^{NC}(2|X, \beta, \Delta) &\equiv \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : \varepsilon_1 \leq X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1, \varepsilon_2 \leq X_2^{ns'} \beta_2 - X_2^{s'} \Delta_2\}, \\ \mathcal{R}_S^{NC}(1|X, \beta, \Delta) &\equiv \mathbb{R}^2 \setminus (\mathcal{R}_S^{NC}(0|X, \beta) \cup \mathcal{R}_S^{NC}(2|X, \beta, \Delta)). \end{aligned}$$

Note that $\mathcal{R}_S^C(0|X, \beta) = \mathcal{R}_S^{NC}(0|X, \beta)$ for any β . Thus,

If players' behavior is cooperative:

$$S = 0 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^C(0|X, \beta_0)\} + 1 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^C(1|X, \beta_0, \Delta_0)\} + 2 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^C(2|X, \beta_0, \Delta_0)\}. \quad (1)$$

If players' behavior is noncooperative:

$$S = 0 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^{NC}(0|X, \beta_0)\} + 1 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^{NC}(1|X, \beta_0, \Delta_0)\} + 2 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^{NC}(2|X, \beta_0, \Delta_0)\}.$$

From here, let

$$\begin{aligned} \mu_S^C(X, \theta) &\equiv F_1(X_1^{ns'} \beta_1) + F_2(X_2^{ns'} \beta_2) - F_{1,2}(X_1^{ns'} \beta_1, X_2^{ns'} \beta_2 | \rho) \\ &\quad + F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2, X_2^{ns'} \beta_2 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2 | \rho), \\ \mu_S^{NC}(X, \theta) &\equiv F_1(X_1^{ns'} \beta_1) + F_2(X_2^{ns'} \beta_2) - F_{1,2}(X_1^{ns'} \beta_1, X_2^{ns'} \beta_2 | \rho) \\ &\quad + F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1, X_2^{ns'} \beta_2 - X_2^{s'} \Delta_2 | \rho). \end{aligned} \quad (2)$$

$E[S|X] = \mu_S^C(X, \theta_0)$ if players' behavior is cooperative, and $E[S|X] = \mu_S^{NC}(X, \theta_0)$ if players' behavior is noncooperative.

2.4 A global behavioral model that nests cooperative and noncooperative behavior

We will embed both of the candidate behavioral models into a *global* model, where the realization of a behavioral selection mechanism ξ determines the behavioral model selected. We denote $\xi = 1$ if players' behavior is cooperative, and $\xi = 2$ if players' behavior is PSNE. The realization of ξ (and therefore the underlying behavioral model) is unobserved by the econometrician, but it is observed by both players prior to playing the game. Our global model allows for players to always cooperate or always play PSNE, but it also allows for cases where players cooperate in some realizations of the game and follow non-cooperative behavior in others. The precise way through which behavior is selected (e.g, whether players observe a signal from “nature” prior to deciding whether to cooperate, etc.) will be irrelevant as long as the mechanism ξ satisfies the exclusion restrictions in Assumption G2. From (1), in our global behavioral model we have,

$$\begin{aligned} S &= \mathbb{1}\{\xi = 1\} \cdot \left[\mathbb{1}\{\varepsilon \in \mathcal{R}_S^C(1|X, \beta_0, \Delta_0)\} + 2 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^C(2|X, \beta_0, \Delta_0)\} \right] \\ &\quad + \mathbb{1}\{\xi = 2\} \cdot \left[\mathbb{1}\{\varepsilon \in \mathcal{R}_S^{NC}(1|X, \beta_0, \Delta_0)\} + 2 \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}_S^{NC}(2|X, \beta_0, \Delta_0)\} \right] \end{aligned}$$

2.4.1 Introducing an instrument

We will assume the existence of an observable Z that controls for the dependence between the behavioral mechanism ξ and players' payoff covariates. We will refer to this control variable as a (behavioral) "instrument".

Assumption G2 (*An exclusion restriction for the behavioral selection mechanism*) $\xi \perp \varepsilon$, and there exists an observable Z such that: (i) at least one element in X is not deterministic conditional on Z , and (ii) $\xi|(X, Z) \sim \xi|Z$. ■

We can weaken $\xi \perp \varepsilon$ to the *conditional* independence restriction $\xi \perp \varepsilon|(X, Z)$ if we assume a parameterization $F_{\varepsilon|X,Z}(\cdot|X, Z, \rho_0)$ for the conditional distribution of $\varepsilon|(X, Z)$. Z can have elements in common with X , but we rule out the case where X is deterministic conditional on Z . Thus, we rule out cases where $X \subseteq Z$. Let $\pi(Z) \equiv \Pr(\xi = 1|Z)$ (the probability of cooperation conditional on Z), which we will leave nonparametrically specified. Group $U \equiv X \cup Z$. By Assumption G2, we have $E[S|U] = \pi(Z) \cdot \mu_S^C(X, \theta_0) + (1 - \pi(Z)) \cdot \mu_S^{NC}(X, \theta_0)$, which simplifies to,

$$E[S|U] = \mu_S^{NC}(X, \theta_0) + \pi(Z) \cdot \Xi_S(X, \theta_0), \quad (3)$$

where $\Xi_S(X, \theta) \equiv \mu_S^C(X, \theta) - \mu_S^{NC}(X, \theta)$. Our goal is to do inference on θ_0 using the semi-parametric convolution (3) when we observe a random sample³ $(Y_i, X_i, Z_i)_{i=1}^n$ produced by our global behavioral model.

2.5 Identifiability of θ_0 in our global behavioral model

Our model predicts the exclusion restriction $E[S|U] = E[S|\mu_S^C(X, \theta_0), \mu_S^{NC}(X, \theta_0), Z]$. We could pursue inference based on this restriction, but the convolution expression in (3) provides additional structure, which can facilitate identification and mitigate the curse of dimensionality vis-a-vis a method based solely on $E[S|U] = E[S|\mu_S^C(X, \theta_0), \mu_S^{NC}(X, \theta_0), Z]$. To exploit (3), we will first construct an estimator for $\pi(Z)$. Identification of $\pi(Z)$ will require ruling out that both candidate behavioral models are observationally equivalent. To this end, consider the following restriction.

Assumption G3 (*Presence of strategic-interaction effects*) For almost every (a.e) realization of Z , we have $\Pr(X_1^{s'}\Delta_{10} + X_2^{s'}\Delta_{20} > 0 | Z) > 0$. ■

Accordingly, from now on our parameter space Θ will be assumed to satisfy,

$$\left. \begin{aligned} \Pr(X_1^{s'}\Delta_1 \geq 0 \text{ and } X_2^{s'}\Delta_2 \geq 0) &= 1 \\ \Pr(X_1^{s'}\Delta_1 + X_2^{s'}\Delta_2 > 0 | Z) &> 0, \text{ a.e } Z \end{aligned} \right\} \forall \Delta \in \Theta. \quad (4)$$

³We only need to observe $(S_i, X_i, Z_i)_{i=1}^n$.

With constant strategic-interaction effects, (4) would be satisfied if $\Delta_p \geq \underline{c} > 0$ for $p = 1, 2$. Without Assumption G3, both candidate behavioral models could be observationally equivalent, making it impossible to identify $\pi(Z)$. Our assumptions allow for $(X_1^s, X_2^s) \in Z$ but, if this is the case, Assumption G3 would preclude (X_1^s, X_2^s) from having a point-mass at zero. If $X_1^s \notin Z$ or $X_2^s \notin Z$, our assumptions allow for (X_1^s, X_2^s) to have a point-mass at zero.

Notation: In what follows, we will often denote the dimension of a vector ξ as $\mathbb{R}^{d_\xi}$.

2.5.1 Identification and estimation of γ_0 from $Pr(S = 0|X)$

Both of our candidate behavioral models produce identical predictions for the event $S = 0$. From Figure 2, the payoff-covariate regions for $S = 0$ are,

$$\mathcal{R}_S^C(0|X, \beta) = \mathcal{R}_S^{NC}(0|X, \beta) \equiv \mathcal{R}_S(0|X, \beta) = \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : \varepsilon_1 \geq X_1^{ns'} \beta_1, \varepsilon_2 \geq X_2^{ns'} \beta_2\}$$

Thus, $Pr(S = 0|U) = Pr(S = 0|X) = Pr(\varepsilon \in \mathcal{R}_S(0|X, \beta_0)|X)$. For $\gamma \equiv (\beta_1, \beta_2, \rho)$ let,

$$\mathbb{P}_0(X, \gamma) \equiv 1 - F_1(X_1^{ns'} \beta_1) - F_2(X_2^{ns'} \beta_2) + F_{1,2}(X_1^{ns'} \beta_1, X_2^{ns'} \beta_2 | \rho).$$

Our global behavioral model predicts $Pr(S = 0|X) = \mathbb{P}_0(X, \gamma_0)$, and the conditional likelihood of $d_0 \equiv \mathbb{1}\{S = 0\}$ given X is,

$$f(d_0|X) = \mathbb{P}_0(X, \gamma_0)^{d_0} \cdot (1 - \mathbb{P}_0(X, \gamma_0))^{1-d_0}. \quad (5)$$

We can study the identification and estimation of γ_0 from (5) under the usual MLE restrictions. To this end, we introduce the following assumption.

Assumption I1 (Existence of $2 + \delta$ moments, an exclusion restriction for (X_1^{ns}, X_2^{ns}) , and a full-rank condition) We have $E[\|X\|^{2+\delta}] < \infty$ for some $\delta > 0$. The true parameter value θ_0 belongs in the interior of the parameter space Θ . For each $p = 1, 2$, the support of X_p^{ns} is not contained in any proper linear subspace of $\mathbb{R}^{d_{X_p^{ns}}}$. For $p, q = \{1, 2\}$, there exists a component $X_{p,j}^{ns} \in X_p^{ns}$ such that $X_{p,j}^{ns} \notin X_q^{ns}$. The elements of the information matrix

$$H_0(\gamma) = E \left[\frac{\partial \log \mathbb{P}_0(X, \gamma)}{\partial \gamma} \cdot \frac{\partial \log \mathbb{P}_0(X, \gamma)}{\partial \gamma}' \right]$$

exist and are continuous functions of γ everywhere on Θ . Furthermore, there exists an open neighborhood of γ_0 where $H_0(\gamma)$ has constant rank, and $H_0(\gamma_0)$ is nonsingular. ■

Local identification of γ_0 through the likelihood (5) follows from Assumption I1 under

standard arguments (e.g, see Rothenberg (1971, Theorem 1)). Note that without the stated exclusion restriction between X_1^{ns} and X_2^{ns} , the full-rank property in Assumption I1 may fail. Henceforth, $\widehat{\gamma}$ will denote the resulting MLE estimator for γ_0 .

2.5.2 Identifiability of Δ_0 under the assumption that players display noncooperative behavior with strictly positive probability

We will maintain the restrictions in Assumption I1, so γ_0 is identified and we will treat it as known. We will show that $\Delta_0 \equiv (\Delta_{10}, \Delta_{20})$ is identifiable under the maintained assumption that players display noncooperative behavior with nonzero positive probability. That is, under the maintained assumption that $Pr(\pi(Z) < 1) > 0$. From (3), we can express

$$S = \mu_S^{NC}(X, \theta_0) + \pi(Z) \cdot \Xi_S(X, \theta_0) + \varepsilon_S, \quad \text{where} \quad E[\varepsilon_S|U] = 0. \quad (6)$$

Thus $E[\varepsilon_S \cdot \Xi_S(X, \theta_0)|Z] = 0$ a.e Z , which yields

$$\pi(Z) \cdot E[\Xi_S(X, \theta_0)^2|Z] = E[(S - \mu_S^{NC}(X, \theta_0)) \cdot \Xi_S(X, \theta_0)|Z] \text{ a.e } Z. \quad (7)$$

The restriction in Assumption G3 implies that $E[\Xi_S(X, \theta_0)^2|Z] > 0$ a.e Z . Thus, from (7),

$$\pi(Z) = \frac{E[(S - \mu_S^{NC}(X, \theta_0)) \cdot \Xi_S(X, \theta_0)|Z]}{E[\Xi_S(X, \theta_0)^2|Z]} \text{ a.e } Z. \quad (8)$$

$E[\Xi_S(X, \theta_0)^2|Z] = 0$ if and only if both candidate behavioral models produce the same predictions almost surely. The purpose of Assumption G3 is to rule this out. From (4), we have $E[\Xi_S(X, \theta)^2|Z] > 0 \forall \theta \in \Theta$, a.e Z . Thus, the following semiparametric weights are well-defined $\forall \theta \in \Theta$, a.e Z ,

$$\pi(Z, \theta) \equiv \frac{E[(S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta)|Z]}{E[\Xi_S(X, \theta)^2|Z]}. \quad (9)$$

For each $\theta \in \Theta$, let

$$\mu_S(U, \theta) \equiv \mu_S^{NC}(X, \theta) + \pi(Z, \theta) \cdot \Xi_S(X, \theta). \quad (10)$$

Then our global behavioral model predicts,

$$E[S|U] = \mu_S(U, \theta_0). \quad (11)$$

Note the importance of ruling out that X is deterministic conditional on Z (see Assumption G2). Otherwise, we could write $X \equiv X(Z)$, $U = Z$ and, for any θ , we would have $\pi(Z, \theta) = (E[S|Z] - \mu_S^{NC}(X(Z), \theta))/\Xi_S(X(Z), \theta)$. From here, the expression in (10) would

simplify to $\mu_S(U, \theta) = E[S|Z] = E[S|U]$. Thus, the conditional moment restriction in (11) would be satisfied by any θ , and it would have no identification power for θ_0 . Now, suppose players cooperate almost surely. In this case, $\pi(Z) = 1$ and $\mu_S(U, \theta_0) = \mu_S^C(X, \theta_0)$ w.p.1. Going back to (2), we can see that the strategic effects $X_1^s \Delta_{10}$ and $X_2^s \Delta_{20}$ shift $\mu_S^C(X, \theta_0)$ only through the *aggregate strategic effect* $X_1^s \Delta_{10} + X_2^s \Delta_{20}$. Thus, if players cooperate almost surely, the strategic parameters $\Delta_0 \equiv (\Delta_{10}, \Delta_{20})$ would not be identifiable unless X_1^s and X_2^s have no elements in common (ruling out constant strategic interaction effects), or unless we reduce the dimensionality of Δ_0 , e.g., by imposing the symmetry condition $\Delta_{10} = \Delta_{20}$. For this reason, let us maintain for now the assumption that players display noncooperative behavior with nonzero probability, so $Pr(\pi(Z) < 1) > 0$. We will come back to the case where players are allowed to cooperate almost surely in Section 2.7.

Assumption I2 (*Strictly positive probability of noncooperative behavior*) $Pr(\pi(Z) < 1) > 0$, so players display noncooperative behavior with strictly positive probability. ■

Taking γ_0 as identified and maintaining Assumption I2 we will study identification and inference for Δ_0 based on the conditional moment restriction (11). We will follow previous work (see, e.g., Bierens (1982), Bierens and Ploberger (1997), Chen and Fan (1999), Dominguez and Lobato (2004), Khan and Tamer (2009) and Andrews and Shi (2013)), and we will propose a conditional-GMM procedure that converts the conditional moments in (11) into an infinite number of unconditional moments, aggregated into a CvM population statistic. In our construction we will consider the type of instrument-function space proposed by Dominguez and Lobato (2004). Group $V \equiv (S, U)$ and let $\varphi_S(V, \theta) \equiv S - \mu_S(U, \theta)$. Note that $E[\varphi_S(V, \theta_0)|U] = 0$ w.p.1. For any $(u, \theta) \in \mathbb{R}^{d_U} \times \Theta$ let $\tau(u, \theta) \equiv E[\varphi_S(V, \theta) \cdot \mathbb{1}\{U \leq u\}]$. From (11), using iterated expectations we must have $\tau(u, \theta_0) = 0 \forall u \in \mathbb{R}^{d_U}$.

From here, Dominguez and Lobato (2004, equation 2) can be invoked to show that $E[\varphi_S(V, \theta)|U] = 0$ a.e. $U \Leftrightarrow \tau(u, \theta) = 0$ for a.e. $u \in \mathbb{R}^{d_U}$, a result that follows from Billingsley (1995, Theorem 16.10iii). We will use the space of instrument functions $\{\mathbb{1}\{U \leq u\} : u \in \text{Supp}(U)\}$ to transform (11) into an infinite collection of unconditional moment restrictions, and we will aggregate them through the population statistic,

$$Q_S(\theta) \equiv \frac{1}{2} \int \tau(u, \theta)^2 dF_U(u) = \frac{1}{2} \cdot E[\tau(U, \theta)^2].$$

$Q_S(\theta) \geq 0 \forall \theta$, and θ_0 is a minimizer of $Q_S(\theta)$, since $Q_S(\theta_0) = 0$. By the arguments in Dominguez and Lobato (2004), any θ such that $Q_S(\theta) = 0$ also satisfies $E[S|U] = \mu_S(U, \theta)$ w.p.1. With γ_0 being identified, we will focus on $Q_S(\gamma_0, \Delta)$. It follows that Δ_0 is *identifiable* if $Q_S(\gamma_0, \Delta) > 0 \forall \Delta \in \Theta : \Delta \neq \Delta_0$, and Δ_0 is *locally identifiable* if this holds $\forall \Delta \in \mathcal{A}$, an open

neighborhood of Δ_0 . We maintain that θ_0 belongs in the interior of Θ , so $\frac{\partial Q_S(\theta_0)}{\partial \Delta} = 0$. From here, local identifiability of Δ_0 can follow if the Hessian $\frac{\partial^2 Q_S(\theta)}{\partial \Delta \partial \Delta'}$ is invertible in an open neighborhood of θ_0 . For $p = 1, 2$, denote $\nabla_p F_{1,2}(\epsilon_1, \epsilon_2 | \rho) \equiv \frac{\partial F_{1,2}(\epsilon_1, \epsilon_2 | \rho)}{\partial \epsilon_p}$ and let,

$$\begin{aligned} H_p^{NC}(X, \theta) &\equiv \nabla_p F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1, X_2^{ns'} \beta_2 - X_2^{s'} \Delta_2 | \rho), \\ H^C(X, \theta) &\equiv \sum_{q=1}^2 \nabla_q F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2, X_2^{ns'} \beta_2 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2 | \rho), \\ J_p(U, \theta) &\equiv \pi(Z, \theta) \cdot H^C(X, \theta) + (1 - \pi(Z, \theta)) \cdot H_p^{NC}(X, \theta). \end{aligned}$$

And let,

$$\dot{\mu}_{S,\Delta}(U, \theta) \equiv \begin{pmatrix} -J_1(U, \theta) \cdot X_1^S + \frac{E[J_1(U, \theta) \cdot \Xi_S(X, \theta) \cdot X_1^S | Z]}{E[\Xi_S(X, \theta)^2 | Z]} \cdot \Xi_S(X, \theta) \\ -J_2(U, \theta) \cdot X_2^S + \frac{E[J_2(U, \theta) \cdot \Xi_S(X, \theta) \cdot X_2^S | Z]}{E[\Xi_S(X, \theta)^2 | Z]} \cdot \Xi_S(X, \theta) \end{pmatrix}$$

In Appendix A (Section A1), we show that,

$$\frac{\partial^2 Q_S(\theta_0)}{\partial \Delta \partial \Delta'} = \int E[\dot{\mu}_{S,\Delta}(U, \theta_0) \cdot \mathbb{1}\{U \leq u\}] \cdot E[\dot{\mu}_{S,\Delta}(U, \theta_0)' \cdot \mathbb{1}\{U \leq u\}] dF_U(u). \quad (12)$$

From here, we can describe sufficient conditions for invertibility of $\frac{\partial^2 Q_S(\theta_0)}{\partial \Delta \partial \Delta'}$.

Assumption I3 (A full rank restriction leading to local identification) Maintain the conditions in Assumptions I1 and I2. In addition, for each $p = 1, 2$, the support of X_p^S is not contained in any proper linear subspace of $\mathbb{R}^{d_{X_p^S}}$. ■

Under Assumption I3, our previous restrictions imply that there exists an open neighborhood $\mathcal{N}$ of Δ_0 such that $\int E[\dot{\mu}_{S,\Delta}(U, \gamma_0, \Delta) \cdot \mathbb{1}\{U \leq u\}] \cdot E[\dot{\mu}_{S,\Delta}(U, \gamma_0, \Delta)' \cdot \mathbb{1}\{U \leq u\}] dF_U(u)$ is invertible for all $\Delta \in \mathcal{N}$, leading to local identification of Δ_0 as a locally unique maximizer of $Q_S(\gamma_0, \Delta)$. Note that Assumption I3 does not require that (X_1^S, X_2^S) have full rank, allowing for $X_1^S = X_2^S$ (e.g, constant strategic effects) as a special case.

Remark 1 Without further restrictions, Assumption I3 cannot be satisfied if players cooperate almost surely. On the other hand, it can be satisfied without further restrictions if players display noncooperative behavior almost surely.

Let us examine Remark 1. If $\pi(Z) = 1$ w.p.1 (players cooperate almost surely), we have $J_p(U, \theta_0) = H^C(X, \theta_0)$ and,

$$\dot{\mu}_{S,\Delta}(U, \theta_0) \equiv \begin{pmatrix} -H^C(X, \theta_0) \cdot X_1^S + \frac{E[H^C(X, \theta_0) \cdot \Xi_S(X, \theta_0) \cdot X_1^S | Z]}{E[\Xi_S(X, \theta_0)^2 | Z]} \cdot \Xi_S(X, \theta_0) \\ -H^C(X, \theta_0) \cdot X_2^S + \frac{E[H^C(X, \theta_0) \cdot \Xi_S(X, \theta_0) \cdot X_2^S | Z]}{E[\Xi_S(X, \theta_0)^2 | Z]} \cdot \Xi_S(X, \theta_0). \end{pmatrix}$$

Satisfying Assumption I3 would require (X_1^S, X_2^S) to have full rank, which cannot be satisfied if X_1^S and X_2^S have elements in common, ruling out the basic case of constant strategic interaction effects. Now suppose players display noncooperative behavior almost surely. In this case, $\pi(Z) = 0$ and $J_p(U, \theta_0) = H_p^{NC}(X, \theta_0)$. Thus,

$$\dot{\mu}_{S,\Delta}(U, \theta_0) \equiv \begin{pmatrix} -H_1^{NC}(X, \theta_0) \cdot X_1^S + \frac{E[H_1^{NC}(X, \theta_0) \cdot \Xi_S(X, \theta_0) \cdot X_1^S | Z]}{E[\Xi_S(X, \theta_0)^2 | Z]} \cdot \Xi_S(X, \theta_0) \\ -H_2^{NC}(X, \theta_0) \cdot X_2^S + \frac{E[H_2^{NC}(X, \theta_0) \cdot \Xi_S(X, \theta_0) \cdot X_2^S | Z]}{E[\Xi_S(X, \theta_0)^2 | Z]} \cdot \Xi_S(X, \theta_0) \end{pmatrix}$$

Assumption I3 can be satisfied even if $X_1^S = X_2^S$ since $Pr(H_1^{NC}(X, \theta_0) \neq H_2^{NC}(X, \theta_0)) > 0$ can follow from our exclusion restrictions between X_1^{NS} and X_2^{NS} . While Assumption I3 leads to local identification of Δ_0 , global identification requires that Δ_0 be the *unique* minimizer of $Q_S(\gamma_0, \Delta)$ over $\Delta \in \Theta$. This will occur if $Pr(\mu_S(U, \theta_0) \neq \mu_S(U, \gamma_0, \Delta)) > 0 \forall \Delta \neq \Delta_0$. To this end, consider the following restriction.

Assumption I4 $\forall \Delta \neq \Delta'$ in Θ and a.e Z , $Pr(\mu_S^{NC}(X, \gamma_0, \Delta) \neq \mu_S^{NC}(X, \gamma_0, \Delta') | Z) > 0$, and $Pr(\mu_S^C(X, \gamma_0, \Delta) - \mu_S^{NC}(X, \gamma_0, \Delta) \neq \mu_S^C(X, \gamma_0, \Delta') - \mu_S^{NC}(X, \gamma_0, \Delta') | Z) > 0$. ■

From (9)-(10), Assumption I4 yields $Pr(\mu_S(U, \theta_0) \neq \mu_S(U, \gamma_0, \Delta)) > 0 \forall \Delta \neq \Delta_0$ in Θ , which implies that Δ_0 is the unique minimizer of $Q_S(\gamma_0, \Delta)$ over $\Delta \in \Theta$. The condition in Assumption I4 can be satisfied if there exist player-specific payoff shifters in X , excluded from Z , with rich enough support.

2.6 An estimator for Δ_0 under the assumption that players display noncooperative behavior with strictly positive probability

Recall that $\widehat{\gamma} \equiv (\widehat{\beta}_1', \widehat{\beta}_2', \widehat{\rho})'$ denotes our MLE estimator for $\gamma_0 \equiv (\beta_{1i}', \beta_{20}', \rho_0')'$. Under the restrictions in Assumption I1, $\widehat{\gamma}$ satisfies the usual MLE linear representation,

$$\widehat{\gamma} = \gamma_0 + \frac{1}{n} \sum_{i=1}^n \psi_\gamma(V_i; \gamma_0) + o_p(n^{-1/2}), \quad (13)$$

where $\psi_\gamma(V_i; \gamma_0)$ is the MLE influence function. Maintaining Assumptions I2-I4, our proposal is to construct a conditional-GMM estimator for Δ_0 that minimizes a sample analog of $Q_S(\gamma_0, \Delta)$. We first construct a kernel-based estimator of the behavioral weight $\pi(Z, \gamma_0, \Delta)$ described in (9). Let $K : \mathbb{R}^{d_Z} \rightarrow \mathbb{R}$ and $h_n \rightarrow 0$ denote a kernel function and a bandwidth sequence respectively (whose properties will be described below). We use

the sample analog estimator for the semiparametric weight $\pi(z, \theta)$,

$$\widehat{\pi}(z, \theta) \equiv \frac{\sum_{i=1}^n (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right)}{\sum_{i=1}^n \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right)}. \quad (14)$$

Our estimator for $\varphi(V, \gamma_0, \Delta)$ is $\widehat{\varphi}_S(V_i, \widehat{\gamma}, \Delta) \equiv S_i - (\mu_S^{NC}(X_i, \widehat{\gamma}, \Delta) + \widehat{\pi}(Z_i, \widehat{\gamma}, \Delta) \cdot \Xi_S(X_i, \widehat{\gamma}, \Delta))$. To construct our conditional-GMM estimator we will pre-specify an inference range $\mathcal{Z} \subseteq \text{Supp}(Z)$ for Z , assumed⁴ to ensure certain uniform asymptotic properties (to be described below) for $\widehat{\pi}(z, \theta)$ over $\mathcal{Z} \times \Theta$. For any $u \in \mathbb{R}^{d_U}$, let $\mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} \equiv \mathbb{1}\{U_i \leq u, Z_i \in \mathcal{Z}\}$ and,

$$\widehat{\tau}_{\mathcal{Z}}(u, \theta) \equiv \frac{1}{n} \sum_{i=1}^n \widehat{\varphi}_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\}.$$

$\widehat{\tau}_{\mathcal{Z}}(u, \theta)$ is an estimator for $\tau_{\mathcal{Z}}(u, \theta) \equiv E[\varphi_S(V, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U \leq u\}]$, which is the version of $\tau(u, \theta)$ when we restrict $Z \in \mathcal{Z}$. Next, let $Q_{S, \mathcal{Z}}(\theta) \equiv \frac{1}{2} \cdot \int \tau_{\mathcal{Z}}(u, \theta)^2 dF_U(u) = \frac{1}{2} \cdot E[\tau_{\mathcal{Z}}(U, \theta)^2]$, which is a restricted version of $Q_S(\theta)$ that integrates out Z over $\mathcal{Z}$. We will characterize Δ_0 as the minimizer of $Q_{S, \mathcal{Z}}(\gamma_0, \Delta)$. Accordingly, our sample objective function is,

$$\widehat{Q}_{S, \mathcal{Z}}(\widehat{\gamma}, \Delta) \equiv \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \widehat{\tau}_{\mathcal{Z}}(U_j, \widehat{\gamma}, \Delta)^2 = \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \left[\frac{1}{n} \sum_{i=1}^n \widehat{\varphi}_S(V_i, \widehat{\gamma}, \Delta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right]^2 \quad (15)$$

our estimator for Δ_0 is given by $\widehat{\Delta} = \underset{\Delta \in \Theta}{\text{argmin}} \widehat{Q}_{S, \mathcal{Z}}(\widehat{\gamma}, \Delta)$. We will describe its asymptotic properties under the following set of restrictions.

Assumption E1 (Bandwidth and kernel restrictions) *Our nonnegative bandwidth sequence satisfies $n^{1/2} \cdot h_n^{2d_Z} \rightarrow \infty$, and for an integer $M \geq 2d_Z + 1$, we have $n^{1/2} \cdot h_n^M \rightarrow 0$. Our kernel $K : \mathbb{R}^{d_Z} \rightarrow \mathbb{R}$ is bias-reducing of order M (for the integer M described above), and is symmetric around zero, with bounded support of the form $[-S, S]$. Our kernel is a function of bounded variation, satisfying $|K(\psi)| \leq \overline{K} \forall \psi \in \mathbb{R}^{d_Z}$, for some $\overline{K} < \infty$. ■*

Next, we describe smoothness restrictions involving some functionals of Z . These are comparable to commonly maintained assumptions in semiparametric models with nonparametric components or “generated regressors”.

Assumption E2 (Smoothness restrictions of some functionals of Z)

(i) *Let M be the integer described in Assumption E1. The following functionals are M -times differentiable with respect to z , with bounded derivatives for all $(z, \theta) \in \mathcal{Z} \times \Theta$: $f_Z(z), \mu_I^{\Xi_S}(z, \theta) \equiv$*

⁴Furthermore, the exclusion restriction $\xi|(X, Z) \sim \xi|Z$ in Assumption G2 only needs to be satisfied over our inference range $\mathcal{Z}$.

$E[(S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta) | Z = z]$, $\mu_{II,2}^{\Xi_S}(z, \theta) \equiv E[\Xi_S(X, \theta)^2 | Z = z]$, $\mu_{III,\theta_\ell}^{\Xi_S}(z, \theta) \equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) | Z = z\right]$, $\mu_{IV,\theta_\ell}^{\Xi_S}(z, \theta) \equiv E\left[\frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) | Z = z\right]$, $\mu_{V,\theta_\ell}^{\Xi_S}(z, \theta) \equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) | Z = z\right]$, and $\mu_{VI,\theta_\ell}^{\Xi_S}(z, \theta) \equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} | Z = z\right]$. These functionals are bounded above in absolute value by some $\bar{C} < \infty$. $f_Z(z)$ and $\mu_{II,2}^{\Xi_S}(z, \theta)$ are bounded below by some $\underline{C} > 0$.

(ii) The following functionals are continuously differentiable with respect to z , with bounded first derivative for all $(z, \theta) \in \mathcal{Z} \times \Theta$: $\delta_{I,\theta_\ell,\theta_j}(z, \theta) \equiv E\left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) | Z = z\right]$, $\delta_{II,\theta_\ell,\theta_j}(z, \theta) \equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_j} | Z = z\right]$, $\delta_{III,\theta_\ell,\theta_j}(z, \theta) \equiv E\left[\frac{\partial^2 \mu_S^{NC}(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) | Z = z\right]$, $\delta_{IV,\theta_\ell,\theta_j}(z, \theta) \equiv E\left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) | Z = z\right]$, and $\delta_{V,\theta_\ell,\theta_j}(z, \theta) \equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} | Z = z\right]$. ■

Finally, we introduce some smoothness and regularity restrictions with respect to θ . These are comparable to commonly maintained assumptions for extremum estimators.

Assumption E3 (Lipschitz-restrictions with respect to θ)

Each of the functions below has the following type of Lipschitz property: $|g(u, \theta) - g(u, \theta')| \leq \bar{G}(u) \cdot \|\theta - \theta'\| \forall u \in \text{Supp}(U)$ (unless noted otherwise), and $\forall \theta, \theta' \in \Theta$, where $\bar{G}(\cdot)$ is a nonnegative function that satisfies $E[\bar{G}(U)^{2+\delta}] < \infty$ for some $\delta > 0$.

(i) The Lipschitz property holds for all $x \in \text{Supp}(X)$ for the following functions: $\mu_S^{NC}(x, \theta)$, $\mu_S^C(x, \theta)$, $\frac{\partial \mu_S^C(x, \theta)}{\partial \theta_\ell}$, $\frac{\partial \mu_S^C(x, \theta)}{\partial \theta_\ell}$, with $\bar{G}(X)$ satisfying $E[\bar{G}(X)^{2+\delta}] < \infty$ for some $\delta > 0$.

(ii) The Lipschitz property holds for all $z \in \mathcal{Z}$ for the following functionals defined in Assumption E2: $\mu_I^{\Xi_S}(z, \theta)$, $\mu_{II,2}^{\Xi_S}(z, \theta)$, $\mu_{III,\theta_\ell}^{\Xi_S}(z, \theta)$, $\mu_{IV,\theta_\ell}^{\Xi_S}(z, \theta)$, $\mu_{V,\theta_\ell}^{\Xi_S}(z, \theta)$, $\mu_{VI,\theta_\ell}^{\Xi_S}(z, \theta)$. In each case, $\bar{G}(Z)$ satisfies $E[\bar{G}(Z)^{2+\delta}] < \infty$ for some $\delta > 0$.

(iii) The Lipschitz property holds for all $z \in \mathcal{Z}$ for the following functionals defined in Assumption E2: $\delta_{I,\theta_\ell,\theta_j}(z, \theta)$, $\delta_{II,\theta_\ell,\theta_j}(z, \theta)$, $\delta_{III,\theta_\ell,\theta_j}(z, \theta)$, $\delta_{IV,\theta_\ell,\theta_j}(z, \theta)$, $\delta_{V,\theta_\ell,\theta_j}(z, \theta)$. In each case, $\bar{G}(Z)$ satisfies $E[\bar{G}(Z)^{2+\delta}] < \infty$ for some $\delta > 0$. ■

Proposition 1 If the restrictions in Assumptions G1-G3, I1-I4 and E1-E3 hold, the estimator $\widehat{\Delta}$ satisfies $\widehat{\Delta} \xrightarrow{p} \Delta_0$, with an asymptotic linear representation of the form,

$$\widehat{\Delta} - \Delta_0 = \frac{1}{n} \sum_{i=1}^n \psi_{\Delta,n}(V_i; \theta_0) + o_p(n^{-1/2}), \quad (16)$$

and $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0) \xrightarrow{d} \mathcal{N}(0, \Omega_\Delta)$, where $\Omega_\Delta = \lim_{n \rightarrow \infty} E[\psi_{\Delta,n}(V; \theta_0) \cdot \psi_{\Delta,n}(V; \theta_0)']$.

Proof of Proposition 1: We outline the steps of the proof below. All the step-by-step details are included in the Econometric Supplement (Section S1). Since $\theta_0 \in \text{int}(\Theta)$ by

assumption, smoothness of our sample objective function $\widehat{Q}_{S,Z}(\theta)$ implies that, with probability approaching one (w.p.a.1), $\widehat{\theta}$ satisfies the first-order conditions $\frac{\partial \widehat{Q}_{S,Z}(\widehat{\theta})}{\partial \Delta} = 0$. From here, a linear approximation yields,

$$0 = \frac{\partial \widehat{Q}_{S,Z}(\widehat{\theta})}{\partial \Delta} = \frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} + \frac{\partial^2 \widehat{Q}_{S,Z}(\widetilde{\theta})}{\partial \Delta \partial \gamma'} \cdot (\widehat{\gamma} - \gamma_0) + \frac{\partial^2 \widehat{Q}_{S,Z}(\widetilde{\theta})}{\partial \Delta \partial \Delta'} \cdot (\widehat{\Delta} - \Delta_0)$$

As usual, $\widetilde{\theta}$ belongs in the line segment connecting $\widehat{\theta}$ and θ_0 and, since $\widehat{\theta} \xrightarrow{p} \theta_0$, it satisfies $\widetilde{\theta} \xrightarrow{p} \theta_0$. Given the uniform convergence result $\sup_{\theta \in \Theta} \left| \frac{\partial^2 \widehat{Q}_{S,Z}(\theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 Q_{S,Z}(\theta)}{\partial \theta \partial \theta'} \right| = o_p(1)$ in equation (S1.44) and the assumption that $\frac{\partial^2 Q_{S,Z}(\theta)}{\partial \Delta \partial \Delta'}$ exists and is a continuous function of θ in an open neighborhood $\mathcal{N}$ of θ_0 (see Assumption I3), it follows that with probability approaching one, we can express,

$$\widehat{\Delta} - \Delta_0 = - \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \Delta'}^{-1} + o_p(1) \right) \times \left[\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} + \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \gamma'} + o_p(1) \right) \cdot (\widehat{\gamma} - \gamma_0) \right]$$

Plugging in the linear representation result of the MLE estimator $\widehat{\gamma}$ in (13) we obtain,

$$\widehat{\Delta} - \Delta_0 = - \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \Delta'}^{-1} + o_p(1) \right) \times \left[\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} + \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \gamma'} + o_p(1) \right) \cdot \left(\frac{1}{n} \sum_{i=1}^n \psi_{\gamma}(V_i; \gamma_0) + o_p(n^{-1/2}) \right) \right] \quad (17)$$

The next step is to show that, under our restrictions, we can obtain a linear representation result for $\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta}$. We have,

$$\begin{aligned} \frac{\partial \widehat{Q}_{S,Z}(\theta)}{\partial \Delta} &= \frac{1}{n} \sum_{j=1}^n \left[\left(\frac{1}{n} \sum_{i=1}^n \widehat{\varphi}_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right) \times \left(\frac{1}{n} \sum_{\ell=1}^n \frac{\partial \widehat{\varphi}_S(V_{\ell}, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_{\ell} \leq U_j\} \right) \right] \\ &= \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_{\ell}, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_{\ell} \leq U_j\} \\ &+ \frac{1}{n} \times \left[\frac{1}{n^2} \sum_{j=1}^n \sum_{i \neq j} \left(\widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} + \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right. \right. \\ &\quad \left. \left. + \widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right) \right] \\ &+ \frac{1}{n^2} \times \left[\frac{1}{n} \sum_{j=1}^n \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \right] \end{aligned} \quad (18)$$

Recall that $\widehat{\varphi}_S(V_i, \theta) \equiv S_i - \mu_S^{NC}(X_i, \theta) - \widehat{\pi}(Z_i, \theta) \cdot \Xi_S(X_i, \theta)$, and $\varphi_S(V_i, \theta) \equiv S_i - \mu_S^{NC}(X_i, \theta) -$

$\pi(Z_i, \theta) \cdot \Xi_S(X_i, \theta)$. Thus,

$$\begin{aligned} \widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta) &= -[\widehat{\pi}(Z_i, \theta) - \pi(Z_i, \theta)] \cdot \Xi_S(X_i, \theta), \\ \frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} &= -\left[\frac{\partial \widehat{\pi}(Z_\ell, \theta)}{\partial \Delta} - \frac{\partial \pi(Z_\ell, \theta)}{\partial \Delta} \right] \cdot \Xi_S(X_\ell, \theta) - [\widehat{\pi}(Z_\ell, \theta) - \pi(Z_\ell, \theta)] \cdot \frac{\partial \Xi_S(X_\ell, \theta)}{\partial \Delta} \end{aligned}$$

For a given (z, θ) , denote $R_b^\pi(z, \theta) \equiv E[\Xi_S(X, \theta)^2 | Z = z] \cdot f_Z(z)$, and let

$$\begin{aligned} \psi_{a,n}^\pi(V_i; z, \theta) &\equiv (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E\left[(S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right)\right], \\ \psi_{b,n}^\pi(U_i; z, \theta) &\equiv \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right) - E\left[\Xi_S(X, \theta)^2 \cdot K\left(\frac{Z - z}{h_n}\right)\right], \\ \psi_n^\pi(V_i; z, \theta) &\equiv \frac{1}{R_b^\pi(z, \theta)} \cdot \psi_{a,n}^\pi(V_i; z, \theta) - \frac{\pi(z, \theta)}{R_b^\pi(z, \theta)} \cdot \psi_{b,n}^\pi(U_i; z, \theta) \end{aligned} \quad (19)$$

In Section S1.1 of the Econometric Supplement we show that $\widehat{\pi}(z, \theta)$ satisfies a linear representation result under the restrictions of Proposition 1,

$$\widehat{\pi}(z, \theta) = \pi(z, \theta) + \frac{1}{n \cdot h_n^{d_z}} \sum_{i=1}^n \psi_n^\pi(V_i; z, \theta) + \vartheta_n^\pi(z, \theta), \quad \text{where} \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\vartheta_n^\pi(z, \theta)| = o_p(n^{-1/2}). \quad (20)$$

By inspection we can verify that $E[\psi_n^\pi(V_i; z, \theta)] = 0 \ \forall (z, \theta) \in \mathcal{Z} \times \Theta$. In the Econometric Supplement we show that $\frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell}$ also satisfies a linear representation result for each θ_ℓ ,

$$\frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} = \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} + \frac{1}{n \cdot h_n^{d_z}} \sum_{i=1}^n \psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta) + \vartheta_n^{\nabla \theta_\ell \pi}(z, \theta), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\vartheta_n^{\nabla \theta_\ell \pi}(z, \theta)| = o_p(n^{-1/2}).$$

The influence function $\psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta)$ is described in the Econometric Supplement (equation S1.22), and it satisfies $E[\psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta)] = 0 \ \forall (z, \theta) \in \mathcal{Z} \times \Theta$. Our next step is to plug in these linear representations into the expression for $\frac{\partial \widehat{Q}_{S,Z}(\theta)}{\partial \Delta}$ in (18). This yields a linear combination of U-statistics of orders 2, 3, and 4. Let $(V_1, V_2, V_3, V_4) \sim F_V \otimes F_V \otimes F_V \otimes F_V$ (four randomly drawn observations from our i.i.d sample $(V_i)_{i=1}^n$) and define,

$$\begin{aligned} \Gamma_{\Delta,n}(V_1; \theta_0) &\equiv \varphi_S(V_1, \theta_0) \cdot E\left[E\left[\frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \middle| U_2\right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \middle| U_1\right] \\ &\quad - E\left[\frac{\partial \varphi_S(V_4, \theta_0)}{\partial \Delta} \cdot E\left[\Xi_S(X_2, \theta_0) \cdot \frac{1}{h_n^{d_z}} \psi_n^\pi(V_1; Z_2, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \middle| V_1, U_3\right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \middle| V_1\right]. \end{aligned} \quad (21)$$

By inspection we can verify that $E[\Gamma_{\Delta,n}(V_1; \theta_0)] = 0$. Analyzing the Hoeffding decompositions (see Serfling (1980, pages 177-178) or Sherman (1994, equations (6)-(7))) and

applying maximal-inequality results for U-processes in Sherman (1994, equations (6)-(7)), we obtain, $\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} = \frac{1}{n} \sum_{i=1}^n \Gamma_{\Delta,n}(V_i; \theta_0) + o_p(n^{-1/2})$. Going back to (17), this yields $\widehat{\Delta} - \Delta_0 = \frac{1}{n} \sum_{i=1}^n \psi_{\Delta,n}(V_i; \theta_0) + o_p(n^{-1/2})$, the result in (16), with

$$\psi_{\Delta,n}(V_i; \theta_0) \equiv -\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \Delta'}^{-1} \times \left(\Gamma_{\Delta,n}(V_i; \theta_0) + \frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \gamma'} \cdot \psi_\gamma(V_i; \gamma_0) \right) \quad (22)$$

This is the influence function in the statement in Proposition 1. ■

In Appendix A we describe an estimator for the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$. Our estimator is of the form $\widehat{\Omega}_\Delta \equiv \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta,n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta,n}(V_i; \widehat{\theta})'$, where $\widehat{\psi}_{\Delta,n}(V_i; \widehat{\theta})$ is an estimator of the influence function $\psi_{\Delta,n}(V_i; \theta_0)$, described in (22).

2.6.1 Estimating the behavioral weight $\pi(z)$

Equipped with $\widehat{\theta} \equiv (\widehat{\gamma}, \widehat{\Delta})$, we can estimate $\pi(z)$, the probability of cooperation conditional on $Z = z$, with $\widehat{\pi}(z, \widehat{\theta})$, where $\widehat{\pi}(z, \theta)$ is as described in (14). The asymptotic properties of our estimated behavioral weights are summarized in (20).

2.6.2 Testing the model

The validity of the instrument Z , and the restrictions leading to Proposition 1 can be tested choosing among multiple available methods for semiparametric models. The goal would be to test “ $H_0 : E[S|U] = \mu_S(U, \theta_0)$ for a.e U ” against “ $H_1 : Pr(E[S|U] \neq \mu_S(U, \theta_0)) > 0$ ”. For instance, following Fan and Li (1996), Zheng (1996) and Aradillas-López (2012, Section 4), we can focus on testing $H_0 : E[\varepsilon_S E[\varepsilon_S|U] f_U(U)] = 0$, where $\varepsilon_S \equiv S - \mu_S(U, \theta_0)$ and $f_U(\cdot)$ is the density function of U . Note that $E[\varepsilon_S E[\varepsilon_S|U] f_U(U)] = E[(E[\varepsilon_S|U])^2 f_U(U)]$, so we would be testing $H_0 : E[(E[\varepsilon_S|U])^2 f_U(U)] = 0$, which would hold if and only if $E[S|U] = \mu_S(U, \theta_0)$ for a.e U . Assuming for exposition⁵ that U is jointly continuously distributed and denoting $\widehat{\varepsilon}_i \equiv S_i - \widehat{\mu}_S(U_i, \widehat{\theta})$, we can use as test-statistic,

$$\frac{1}{n \cdot (n-1) \cdot b_n^{d_U}} \sum_{i=1}^n \sum_{j \neq i} \widehat{\varepsilon}_i \cdot \widehat{\varepsilon}_j \cdot \mathcal{K}\left(\frac{U_i - U_j}{b_n}\right),$$

where $b_n \rightarrow 0$ is a bandwidth sequence and $\mathcal{K} : \mathbb{R}^{d_U} \rightarrow \mathbb{R}$ is a kernel function. The asymptotic properties of this type of test-statistic and of the resulting test can be obtained following the steps in Aradillas-López (2012, Section 4), which studies models with generated regressors with asymptotic properties like those described in (20) for $\widehat{\pi}(z, \widehat{\theta})$.

⁵This type of test can be readily modifiable when U includes continuous and also discrete covariates.

2.7 Inference for Δ_0 when we allow for the possibility that players cooperate almost surely

As we showed above, allowing for the possibility that $\pi(Z) = 1$ w.p.1 implies that Δ_0 may not be identified without further restrictions. In this case, our suggestion is to use (11) to construct a confidence set (CS) for Δ_0 . Maintaining Assumption I1, γ_0 is identified and estimable using MLE. Plugging in $\widehat{\gamma}$, a CS for Δ_0 can be obtained using (11). To this end, a number of existing methods based on conditional moment equalities can be applied. Adapting an existing method that facilitates the use of the first-step estimator $\widehat{\gamma}$ and the generated regressors $\widehat{\pi}(z, \theta)$ can be done using standard arguments and is relatively straightforward. For this reason, we omit the details for brevity.

3 Allowing for mixed strategies

Let us maintain the two candidate behavioral models as cooperation (as described previously) and Nash equilibrium. Allowing for mixed strategies does not change the predictions of our cooperative behavioral model since the optimal choice in that case is unique w.p.1. In contrast, our noncooperative behavioral model would now have three Nash equilibria (NE) in the region of multiplicity. Namely, two pure-strategy NE (PSNE), and one mixed-strategy NE (MSNE). The possibility of selecting the MSNE means that $S \equiv Y_1 + Y_2$ is no longer uniquely predicted by our noncooperative model. Thus, instead of S , we will focus now directly on the predictions for the outcome Y made by each of our candidate behavioral models. We will also have to make assumptions about the equilibrium selection mechanism in the NE behavioral model. For the cooperative model let,

$$\mathcal{R}^C(y|X, \theta_0) \equiv \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : y \text{ is the optimal outcome under cooperation, given } (X, \theta_0)\}.$$

Next, for the noncooperative model let,

$$\mathcal{U}^{NE}(y|X, \theta_0) \equiv \{(\varepsilon_1, \varepsilon_2) \in \mathbb{R}^2 : y \text{ is the unique NE of the game, given } (X, \theta_0)\}$$

All the above regions can be obtained from Figure 1, where we can also obtain $\mathcal{M}^{NE}(X, \theta_0)$, the multiple NE region. When $\varepsilon \in \mathcal{M}^{NE}(X, \theta_0)$, the game has two pure-strategy NE (PSNE), $(1, 0)$ and $(0, 1)$, and one mixed-strategy NE (MSNE). We will index the multiple NE as

- NE #1: PSNE $(1, 0)$, • NE #2: PSNE $(0, 1)$, • NE #3: MSNE.

Let $Y_{(\ell)}$ denote the (potential) outcome that would be produced by NE $\ell \in \{1, 2, 3\}$ in the multiple NE region $\mathcal{M}^{NE}(X, \theta_0)$. Denoting $Pr(Y_{(\ell)} = y|X, \varepsilon) \equiv \sigma_{\mathcal{M}, \ell}^{NE}(y|X, \varepsilon, \theta_0)$ (the distribu-

tion of $Y|(X, \varepsilon)$ induced by NE ℓ) and solving for the MSNE, we have

- For NE #1 (PSNE (1, 0)): $\sigma_{\mathcal{M},1}^{NE}(y|X, \varepsilon, \theta_0) = \mathbb{1}\{y = (1, 0)\}$,
- For NE #2 (PSNE (0, 1)): $\sigma_{\mathcal{M},2}^{NE}(y|X, \varepsilon, \theta_0) = \mathbb{1}\{y = (0, 1)\}$,
- For NE #3 (MSNE):

$$\sigma_{\mathcal{M},3}^{NE}(y|X, \varepsilon, \theta_0) = \left(\frac{X_2^{ns'}\beta_{20} - \varepsilon_2}{X_2^{s'}\Delta_{20}} \right)^{y_1} \left(1 - \frac{X_2^{ns'}\beta_{20} - \varepsilon_2}{X_2^{s'}\Delta_{20}} \right)^{1-y_1} \left(\frac{X_1^{ns'}\beta_{10} - \varepsilon_1}{X_1^{s'}\Delta_{10}} \right)^{y_2} \left(1 - \frac{X_1^{ns'}\beta_{10} - \varepsilon_1}{X_1^{s'}\Delta_{10}} \right)^{1-y_2} \quad (23)$$

3.1 Behavioral and equilibrium selection mechanisms

Again, denote our behavioral selection mechanism by ξ , with $\xi = 1$ if players' behavior is cooperative, and $\xi = 2$ if players play NE. Allowing for mixed strategies means that we now have to explicitly introduce an *equilibrium selection mechanism* that chooses the NE to be played when $\xi = 2$ and $\varepsilon \in \mathcal{M}^{NE}(X, \theta_0)$. Let λ denote our NE equilibrium selection mechanism. Then $\lambda \in \{1, 2, 3\}$, with $\lambda = \ell$ indicating the selection of NE ℓ when $\xi = 2$ and $\varepsilon \in \mathcal{M}^{NE}(X, \theta_0)$. For each $y \in \mathcal{Y}$, our global behavioral model yields,

$$\begin{aligned} \mathbb{1}\{Y = y\} = & \mathbb{1}\{\xi = 1\} \cdot \mathbb{1}\{\varepsilon \in \mathcal{R}^C(y|X, \beta_0, \Delta_0)\} \\ & + \mathbb{1}\{\xi = 2\} \cdot \left(\mathbb{1}\{\varepsilon \in \mathcal{U}^{NE}(y|X, \beta_0, \Delta_0)\} + \mathbb{1}\{\varepsilon \in \mathcal{M}^{NE}(X, \beta_0, \Delta_0)\} \cdot \sum_{\ell=1}^3 \mathbb{1}\{\lambda = \ell\} \cdot \mathbb{1}\{Y_{(\ell)}^{NE} = y\} \right). \end{aligned}$$

The prediction for $Pr(Y = y|X)$ when players cooperate is,

$$p^C(y|X, \theta_0) \equiv \int \mathbb{1}\{\varepsilon \in \mathcal{R}^C(y|X, \beta_0, \Delta_0)\} f_{1,2}(\varepsilon|\rho_0) d\varepsilon. \quad (24)$$

Next, let

$$\begin{aligned} g_{\mathcal{U}}^{NE}(y|X, \theta_0) & \equiv \int \mathbb{1}\{\varepsilon \in \mathcal{U}^{NE}(y|X, \beta_0, \Delta_0)\} f_{1,2}(\varepsilon|\rho_0) d\varepsilon, \\ g_{\mathcal{M},\ell}^{NE}(y|X, \theta_0) & \equiv \int \sigma_{\mathcal{M},\ell}^{NE}(y|X, \varepsilon, \beta_0, \Delta_0) \cdot \mathbb{1}\{\varepsilon \in \mathcal{M}^{NE}(X, \beta_0, \Delta_0)\} f_{1,2}(\varepsilon|\rho_0) d\varepsilon. \end{aligned}$$

The prediction for $Pr(Y = y|X)$ with noncooperative behavior when NE ℓ is selected is,

$$p_{\ell}^{NC}(y|X, \theta_0) \equiv g_{\mathcal{U}}^{NE}(y|X, \theta_0) + g_{\mathcal{M},\ell}^{NE}(y|X, \theta_0). \quad (25)$$

To extend our previous approach to this expanded model we will assume that the observable instrument Z has the same type of control function properties as before, now with respect to both the behavioral and the equilibrium selection mechanisms.

Assumption E4 (*Exclusion restrictions with cooperative and NE behavior with mixed strategies*) $\xi \perp \varepsilon$, and there exists an observable Z such that: (i) at least one element in X is not deterministic conditional on Z , (ii) $(\xi, \lambda)|(X, Z) \sim (\xi, \lambda)|Z$, and (iii) conditional on $\varepsilon \in \mathcal{M}^{NE}(X, \beta_0, \Delta_0)$, we have $\lambda|(\xi, \varepsilon, X, Z) \sim \lambda|(\xi, Z)$. ■

The restriction $\xi \perp \varepsilon$ can be relaxed to $\xi \perp \varepsilon|(X, Z)$ if we parameterize the conditional distribution of $\varepsilon|(X, Z)$. We denote $Pr(\xi = 1|Z) \equiv \pi(Z)$ and $Pr(\lambda = \ell|\xi = 2, Z) \equiv \omega(\ell|Z)$, where $\sum_{\ell=1}^3 \omega(\ell|Z) = 1$. We will leave $\pi(Z)$ and $(\omega(\ell|Z))_{\ell=1}^3$ nonparametrically specified. Let,

$$\begin{aligned} m_1(y|X, \theta_0) &\equiv p^C(y|X, \theta_0) - p_3^{NC}(y|X, \theta_0), \\ m_2(y|X, \theta_0) &\equiv p_3^{NC}(y|X, \theta_0) - p_1^{NC}(y|X, \theta_0), \\ m_3(y|X, \theta_0) &\equiv p_3^{NC}(y|X, \theta_0) - p_2^{NC}(y|X, \theta_0). \end{aligned} \quad (26)$$

The functions in (26) measure the difference in the predictions for $Pr(Y = y|X)$ of each candidate solution relative to NE $\ell = 3$. In what follows, any other candidate solution could be taken as the reference. Letting $\vartheta(\ell|Z) \equiv (1 - \pi(Z)) \cdot \omega(\ell|Z)$, and using the fact that $\omega(3|Z) = 1 - \omega(1|Z) - \omega(2|Z)$, Assumption E4 yields,

$$p_Y(y|U) = p_3^{NC}(y|X, \theta_0) + \pi(Z) \cdot m_1(y|X, \theta_0) + \vartheta(1|Z) \cdot m_2(y|X, \theta_0) + \vartheta(2|Z) \cdot m_3(y|X, \theta_0). \quad (27)$$

Grouping

$$\delta(Z) \equiv (\pi(Z), \vartheta(1|Z), \vartheta(2|Z))', \quad \Xi(y|X, \theta_0) \equiv (m_1(y|X, \theta_0), m_2(y|X, \theta_0), m_3(y|X, \theta_0))',$$

the above can be expressed compactly as,

$$p_Y(y|U) = p_3^{NC}(y|X, \theta_0) + \delta(Z)' \Xi(y|X, \theta_0). \quad (28)$$

This is the semiparametric behavioral convolution when we allow for mixed strategies.

3.2 Identifiability and estimation of θ_0 with mixed strategies

If players cooperate almost surely, we have $p_Y(y|U) = p^C(y|X, \theta_0)$ and, by the arguments in previous sections, Δ_0 is not identifiable without further restrictions. Let us maintain Assumption I2, so players are assumed to follow noncooperative behavior with strictly

positive probability. Our goal is to use the semiparametric convolution in (28) to do inference on θ_0 . Allowing for mixed-strategies means that the subvector of parameters γ_0 can no longer be identified from $Pr(Y_1 + Y_2 = 0|X)$. Thus, we will focus on using (28) to do inference on the entire vector of parameters θ_0 . Our first step is to describe conditions that lead to identification of the behavioral weights $\delta(Z)$. Identification will be made possible by the following result.

Result M *Under the conditions in Assumption G3 and the restrictions to the parameter space Θ in (4), $E[\Xi(y|X, \theta) \cdot \Xi(y|X, \theta)'|Z]$ is invertible for every $\theta \in \Theta$ and a.e Z for $y \in \{(1, 0), (0, 1)\}$.*

Appendix A2 contains the proof of Result M. In essence, identification of $\delta(Z)$ requires the existence of at least one outcome $y \in \mathcal{Y}$ such that, for a.e Z , and any possible value that θ_0 can take in Θ , the following holds:

- (i) *Across behavioral models.*- The predictions for $Pr(Y = y|X)$ for the cooperative and noncooperative behavioral models are different with nonzero probability (with respect to X) for every possible solution (i.e, every NE) of the noncooperative behavioral model.
- (ii) *Within behavioral models.*- The predictions for $Pr(Y = y|X)$ for any pair of possible co-existing solutions (i.e, any pair of co-existing NE) in the noncooperative behavioral are different with nonzero probability.

Result M states that the above requirements are satisfied for $y \in \{(1, 0), (0, 1)\}$ under our restrictions. Letting $D(y) \equiv \mathbb{1}\{Y = y\}$ and fixing $\bar{y} \in \{(1, 0), (0, 1)\}$, Result M allows us to characterize $\delta(Z)$ as

$$\delta(Z) = E[\Xi(\bar{y}|X, \theta_0) \cdot \Xi(\bar{y}|X, \theta_0)'|Z]^{-1} \cdot E[\Xi(\bar{y}|X, \theta_0) \cdot (D(\bar{y}) - p_3^{NC}(\bar{y}|X, \theta_0))|Z],$$

a.e Z , which is analogous to (8) in our pure-strategies model. Result M also implies that the following weights are well-defined for any $\theta \in \Theta$ and a.e Z ,

$$\delta(Z, \theta) \equiv E[\Xi(\bar{y}|X, \theta) \cdot \Xi(\bar{y}|X, \theta)'|Z]^{-1} \cdot E[\Xi(\bar{y}|X, \theta) \cdot (D(\bar{y}) - p_3^{NC}(\bar{y}|X, \theta))|Z]$$

From here let,

$$\mathbb{P}(y|U, \theta) = p_3^{NC}(y|X, \theta) + \delta(Z, \theta)' \Xi(y|X, \theta). \quad (29)$$

Then (28) yields,

$$p_Y(y|U) = \mathbb{P}(y|U, \theta_0) \quad \forall y \in \mathcal{Y}, \text{ a.e } U. \quad (30)$$

Since $\sum_{y \in \mathcal{Y}} p_Y(y|U) = 1$, we can focus on any $|\mathcal{Y}| - 1 (= 3)$ outcomes. Let us focus on $\mathcal{Y}^* \equiv \{(1, 0), (0, 1), (1, 1)\}$. From (29), θ is observationally equivalent to θ_0 if $\mathbb{P}(y|U, \theta) =$

$\mathbb{P}(y|U, \theta)$ w.p.1 for each $y \in \mathcal{Y}^*$. Therefore, θ_0 is identifiable from (30) if, $\exists y \in \mathcal{Y}^*$ such that, $Pr(\mathbb{P}(y|U, \theta) \neq \mathbb{P}(y|U, \theta_0)) > 0 \forall \theta \in \Theta: \theta \neq \theta_0$, and θ_0 is locally identifiable if there exists a neighborhood $\mathcal{A}$ of θ_0 where this condition holds. In the spirit of our previous sections, we suggest a conditional GMM approach based on Dominguez and Lobato (2004). Group $V \equiv (Y, U)$. For $(y, \theta, u) \in \mathcal{Y}^* \times \Theta \times \mathbb{R}^{d_U}$, let

$$\varphi(y|V, \theta) \equiv D(y) - \mathbb{P}(y|U, \theta) \quad \text{and} \quad T(y|u, \theta) \equiv E[\varphi(y|V, \theta) \cdot \mathbb{1}\{U \leq u\}].$$

From (30) and iterated expectations, $T(y|u, \theta_0) = 0 \forall u \in \mathbb{R}^{d_U}, y \in \mathcal{Y}^*$. Let

$$Q_Y(y|\theta) \equiv \frac{1}{2} \int T(y|u, \theta)^2 dF_U(u) = \frac{1}{2} E[T(y|U, \theta)^2].$$

$Q_Y(y|\theta) \geq 0 \forall \theta$ and, from the arguments in Dominguez and Lobato (2004), we have $Q_Y(y|\theta) = 0 \Leftrightarrow T(y|u, \theta) = 0$ F_U -a.e $u \in \mathbb{R}^{d_U}$. We can aggregate over $y \in \mathcal{Y}^*$ through,

$$Q_Y(\theta) \equiv \sum_{y \in \mathcal{Y}^*} Q_Y(y|\theta).$$

We can then characterize θ_0 as a minimizer of $Q_Y(\theta)$ over $\theta \in \Theta$. The following restrictions are sufficient for local identification of θ_0 .

Assumption I5 *The true parameter value θ_0 belongs in the interior of the parameter space Θ . In addition, the following conditions hold.*

- (i) *For each $p = 1, 2$, the support of X_p^{ns} is not contained in any proper linear subspace of $\mathbb{R}^{d_{X_p^{ns}}}$. For $p, q = \{1, 2\}$, there exists a component $X_{p,j}^{ns} \in X_p^{ns}$ such that $X_{p,j}^{ns} \notin X_q^{ns}$.*
- (ii) *For each $p = 1, 2$, the support of X_p^s is not contained in any proper linear subspace of $\mathbb{R}^{d_{X_p^s}}$.*

Similar arguments to those following Assumption I3 can be used to show that there exists an open neighborhood $\mathcal{A} \subseteq \Theta$ of θ_0 such that $\frac{\partial^2 Q_Y(\theta)}{\partial \theta \partial \theta'}$ is invertible $\forall \theta \in \mathcal{A}$, and local identification of θ_0 would follow. Also similar to the pure-strategy case, the availability of player-specific payoff shifters in X with rich enough support, and excluded from Z , would lead to global identification of θ_0 . In this case, for $y \in \{(1, 0), (0, 1)\}$, any $\theta \neq \theta_0$ would satisfy the following for a.e Z ,

- $Pr(p^C(y|X, \theta) \neq p^C(y|X, \theta_0)|Z) > 0$,
- $Pr(g_{\mathcal{M}, \ell}^{NE}(y|X, \theta) \neq g_{\mathcal{M}, \ell}^{NE}(y|X, \theta_0)|Z) > 0$ for $\ell = 1, 2, 3$,
- $Pr(g_{\mathcal{U}}^{NE}(y|X, \theta) \neq g_{\mathcal{U}}^{NE}(y|X, \theta_0)|Z) > 0$.

From here we would have $Pr(\mathbb{P}(y|U, \theta) \neq \mathbb{P}(y|U, \theta_0)|Z) > 0 \forall \theta \neq \theta_0$, a.e Z and θ_0 would be globally identified as the unique minimizer of $Q_Y(\theta)$ over Θ .

3.2.1 An estimator for θ_0 with mixed strategies

As before, our approach would begin with the construction of a semiparametric estimator for the behavioral weights $\delta(Z, \theta)$. For each (z, θ) define,

$$\widehat{\delta}(z, \theta) \equiv \left(\sum_{i=1}^n \Xi(\bar{y}|X_i, \theta) \cdot \Xi(\bar{y}|X_i, \theta)' K\left(\frac{Z_i - z}{h_n}\right) \right)^{-1} \sum_{i=1}^n \Xi(\bar{y}|X_i, \theta) \cdot (D_i(\bar{y}) - p_3^{NC}(\bar{y}|X_i, \theta)) K\left(\frac{Z_i - z}{h_n}\right),$$

$$\widehat{\mathbb{P}}(y|U, \theta) \equiv p_3^{NC}(y|X, \theta) + \widehat{\delta}(Z, \theta)' \Xi(y|X, \theta), \quad \widehat{\varphi}(y|V, \theta) \equiv D(y) - \widehat{\mathbb{P}}(y|U, \theta).$$

As in the pure-strategy case, we pre-specify an inference range $\mathcal{Z} \subseteq \text{Supp}(Z)$ where our weights satisfy uniform asymptotic properties. For any $u \in \mathbb{R}^{d_U}$, let $\mathbb{1}_{\mathcal{Z}}\{U \leq u\} \equiv \mathbb{1}\{U \leq u, Z \in \mathcal{Z}\}$. We will define $T_{\mathcal{Z}}(y|u, \theta) \equiv E[\varphi(y|V, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U \leq u\}]$, the version of $T(y|u, \theta)$ where Z is restricted to $\mathcal{Z}$. Next, let

$$Q_{Y, \mathcal{Z}}(y|\theta) \equiv \frac{1}{2} \int T_{\mathcal{Z}}(y|u, \theta)^2 dF_U(u) = \frac{1}{2} E[T_{\mathcal{Z}}(y|U, \theta)^2], \quad \text{and} \quad Q_{Y, \mathcal{Z}}(\theta) \equiv \sum_{y \in \mathcal{Y}^*} Q_{Y, \mathcal{Z}}(y|\theta).$$

$Q_{Y, \mathcal{Z}}(\theta)$ is our population statistic restricted to $\mathcal{Z}$. Our sample objective function is,

$$\widehat{Q}_{Y, \mathcal{Z}}(\theta) \equiv \sum_{y \in \mathcal{Y}^*} \widehat{Q}_{Y, \mathcal{Z}}(y|\theta), \quad \text{where}$$

$$\widehat{Q}_{Y, \mathcal{Z}}(y|\theta) \equiv \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \widehat{T}_{\mathcal{Z}}(y|U_j, \theta)^2 = \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \left[\frac{1}{n} \sum_{i=1}^n \widehat{\varphi}(y|V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right]^2,$$

Our proposed estimator for θ_0 is $\widehat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \widehat{Q}_{Y, \mathcal{Z}}(\theta)$. Under Assumptions G3, I2, I5, E1, E4, and the appropriate modification of Assumptions E2 and E3, our estimator $\widehat{\theta}$ is $\sqrt{n}$ -consistent and asymptotically normal. This is shown following analogous steps to the proof of Proposition 1, minus the presence of the first-stage estimator $\widehat{\gamma}$. Let us sketch the arguments next. Let $(V_1, V_2, V_3, V_4) \sim F_V \otimes F_V \otimes F_V \otimes F_V$ (four randomly drawn observations from our i.i.d sample $(V_i)_{i=1}^n$) and define,

$$\Gamma_{\theta, n}(y|V_1; \theta_0) \equiv E \left[E \left[\frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \middle| U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \middle| U_1 \right] \cdot \varphi(y|V_1, \theta_0)$$

$$- E \left[\frac{\partial \varphi(y|V_4, \theta)}{\partial \theta} \cdot E \left[\frac{1}{h_n^{d_Z}} \psi_n^\delta(V_1; Z_2, \theta)' \Xi(y|X_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \middle| V_1, U_3 \right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \middle| V_1 \right].$$

We can verify that $E[\Gamma_{\theta,n}(y|V_1; \theta_0)] = 0$. Let $\Gamma_{\theta,n}(V_i; \theta_0) \equiv \sum_{y \in \mathcal{Y}^*} \Gamma_{\theta,n}(y|V_i; \theta_0)$ and,

$$\psi_{\theta,n}(V_i; \theta_0) \equiv -\frac{\partial^2 Q_{Y,Z}(\theta_0)^{-1}}{\partial \theta \partial \theta'} \times \Gamma_{\theta,n}(V_i; \theta_0).$$

Note that $E[\psi_{\theta,n}(V; \theta_0)] = 0$. In Section S2 of the Econometric Supplement we show that, under Assumptions G3, I2, I5, E1, E4, and the appropriate modification of Assumptions E2 and E3,

$$\widehat{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^n \psi_{\theta,n}(V_i; \theta_0) + o_p(n^{-1/2}). \quad (31)$$

This is an extension of Proposition 1 to the model with mixed strategies. The behavioral weights $\delta(z)$ can be estimated through $\widehat{\delta}(z, \widehat{\theta})$, with asymptotic properties analogous to those described in Section 2.6.1 for $\widehat{\pi}(z, \widehat{\theta})$. Equipped with $\widehat{\delta}(z, \widehat{\theta})$ we can learn about the propensity of behavior selection (cooperative vs. noncooperative) and the propensity of NE selection in the noncooperative behavioral model. A specification test can be conducted using a number of existing semiparametric methods. Our goal would be to test “ $H_0 : p_Y(y|U) = \mathbb{P}(y|U, \theta_0) \forall y \in \mathcal{Y}^*, \text{ a.e } U$ ” against “ $H_1 : Pr(p_Y(y|U) \neq \mathbb{P}(y|U, \theta_0)) > 0$ for some $y \in \mathcal{Y}^*$ ”. We can apply the type of specification test outlined in Section 2.6.2.

3.3 Inference for θ_0 when we allow for the possibility that players cooperate almost surely

If we are unwilling to assume that players behave noncooperatively with nonzero probability, Δ_0 (and thus, θ_0) would not be identifiable from (30) unless we impose additional restrictions such as $\Delta_{10} = \Delta_{20}$. In this case, our proposal would be to construct a CS for θ_0 based on (30). This can be done by adapting an existing method that facilitates the use of the generated regressors $\widehat{\delta}(z, \theta)$. Since this can be carried out using relatively standard arguments, we omit the details for brevity.

4 A general model

Let us go beyond a 2×2 game and consider a general discrete game with a collection of candidate behavioral models, each with possible multiple solutions.

4.1 A parameterized normal-form game

We have a collection of players, $p = 1, \dots, P$, where each p has a discrete and finite action space $\mathcal{Y}_p$ and a parameterized payoff function $u_p(y; X_p, \varepsilon_p, \beta_{p_0})$, where $y \equiv (y_1, \dots, y_P)$ is a particular action profile, $X_p \in \mathbb{R}^{d_{X_p}}$ and $\varepsilon_p \in \mathbb{R}$ denote observable and unobservable (to

the econometrician) payoff shifters, and where β_{p_0} is a finite-dimensional parameter. We denote the outcome of the game as $Y \equiv (Y_1, \dots, Y_P) \in \mathcal{Y}$, where $\mathcal{Y} \equiv \mathcal{Y}_1 \times \dots \times \mathcal{Y}_P$. Group $X \equiv \cup_{p=1}^P X_p \in \mathbb{R}^{d_X}$ and $\varepsilon \equiv (\varepsilon_1, \dots, \varepsilon_P) \in \mathbb{R}^P$. We maintain that $\varepsilon \perp X$ and $\varepsilon \sim F_\varepsilon(\cdot | \rho_0)$, a parametric joint distribution⁶ indexed by a finite dimensional parameter ρ_0 . The parameters of the model are $\theta_0 \equiv (\beta_{1_0}, \dots, \beta_{P_0}, \rho_0) \in \Theta$.

4.2 A collection of candidate behavioral models and solutions

We have a collection of $\mathcal{B}$ *candidate behavioral models*, labeled $b = 1, \dots, \mathcal{B}$. The outcome of each observation of the game is the realization of a *solution* to one of the candidate behavioral models. The econometrician does not know the behavioral model that produced each observation, and different observations in the data could have been produced by different behavioral models. As before, the behavior-selection mechanism is denoted as ξ .

4.2.1 Solutions for each behavioral model

For any given realization of (X, ε) , each behavioral model b predicts a collection of solutions (e.g, Nash equilibria). Assume that each behavioral model has a finite number of solutions for a.e (X, ε) . Behavioral model b comes equipped with a solution selection mechanism λ^b which, for any realization of (X, ε) , describes which solution will be selected among the existing ones. Thus, λ^b selects a particular *solution configuration* across all regions of realizations of payoff shifters. Letting S_b denote the total number of possible solution configurations, we then have $\lambda^b \in \{1, \dots, S_b\}$, and $\lambda^b = \ell$ indicates that the ℓ^{th} solution configuration has been selected.

4.2.2 Predictions for each solution configuration

Let $g_\ell^b(y|X, \theta_0)$ denote the prediction for $Pr(Y = y|X)$ when solution configuration ℓ is selected in behavioral model b . Each $g_\ell^b(y|X, \theta_0)$ has an analytical expression derived from our parameterization of payoffs and of the distribution of unobserved payoff shifters. For example, in the model of Section 3 these analytical expressions are given in (24) and (25).

4.3 Exclusion restriction for an instrument Z

The inference strategy in this paper relies on the availability of an observable instrument Z capable of controlling for the dependence between payoffs and the underlying behavioral and solution selection mechanisms. A general version of the assumptions made in

⁶We can relax the assumption that $\varepsilon \perp X$ and parameterize the conditional distribution of $\varepsilon|X$.

the previous sections can look as follows. Suppose that⁷ $\varepsilon \perp \xi$, and that there exists an observable Z such that: (i) at least one element in X is not deterministic conditional on Z , (ii) $(\xi, \lambda^1, \dots, \lambda^B)|(X, Z) \sim (\xi, \lambda^1, \dots, \lambda^B)|Z$, and⁸ (iii) $\lambda_b|(\xi = b, \varepsilon, X, Z) \sim \lambda_b|(\xi = b, Z)$ for each b . Denote $Pr(\xi = b|Z) \equiv \pi^b(Z)$ and $Pr(\lambda^b = \ell|\xi = b, Z) \equiv \omega^b(\ell|Z)$. These functionals will be left nonparametrically specified.

4.4 An expression for $Pr(Y = y|U)$

Define $\vartheta^b(\ell|Z) \equiv \pi^b(Z) \cdot \omega^b(\ell|Z)$. Fix any behavioral model b_0 and, for each behavioral model b , fix any solution configuration ℓ_0^b . We will use b_0 as our reference behavioral model, and ℓ_0^b as the reference solution configuration for behavioral model b . Group $U \equiv X \cup Z$ and let $p_Y(y|U) \equiv Pr(Y = y|U)$. Using the fact that $\pi^{b_0}(Z) = 1 - \sum_{b \neq b_0} \pi^b(Z)$ and $\omega^b(\ell_0^b|Z) = 1 - \sum_{\ell \neq \ell_0^b} \omega^b(\ell|Z)$, our general model yields

$$p_Y(Y = y|U) = \underbrace{g_{\ell_0^{b_0}}^{b_0}(y|X, \theta_0)}_{\text{reference solution}} + \underbrace{\sum_{b \neq b_0} \pi^b(Z) \cdot (g_{\ell_0^b}^b(y|X, \theta_0) - g_{\ell_0^{b_0}}^{b_0}(y|X, \theta_0))}_{\text{comparisons across behavioral models}} + \underbrace{\sum_{b=1}^B \sum_{\ell \neq \ell_0^b} \vartheta^b(\ell|Z) \cdot (g_{\ell}^b(y|X, \theta_0) - g_{\ell_0^b}^b(y|X, \theta_0))}_{\text{comparisons within behavioral models}}$$

Note that (27) is a special case of the above expression. Group

$$\begin{aligned} \delta(Z) &\equiv \left(\left(\pi^b(Z) \right)_{b \neq b_0}, \left(\left(\vartheta^b(\ell|Z) \right)_{\ell \neq \ell_0^b} \right)_{b=1}^B \right), \\ \Xi(y|X, \theta_0) &\equiv \left(\left(g_{\ell_0^b}^b(y|X, \theta_0) - g_{\ell_0^{b_0}}^{b_0}(y|X, \theta_0) \right)_{b \neq b_0}, \left(\left(g_{\ell}^b(y|X, \theta_0) - g_{\ell_0^b}^b(y|X, \theta_0) \right)_{\ell \neq \ell_0^b} \right)_{b=1}^B \right) \end{aligned} \quad (32)$$

We can then write $p_Y(Y = y|U)$ as a generalization of our expressions in (3) and (28),

$$p_Y(Y = y|U) = g_{\ell_0^{b_0}}^{b_0}(y|X, \theta_0) + \delta(Z)' \Xi(y|X, \theta_0). \quad (33)$$

This is a general version of the semiparametric behavioral convolution that characterized our previous examples.

4.5 Estimation and inference for θ_0

As is apparent from (32), identification of the weights $\delta(Z)$ in (33) requires the existence of at least one outcome $y \in \mathcal{Y}$ such that, for a.e realization of Z , and any possible value that θ_0 can take in Θ , the following must hold:

⁷The restriction $\xi \perp \varepsilon$ can be relaxed to $\xi \perp \varepsilon|(X, Z)$ if we parameterize the conditional distribution of $\varepsilon|(X, Z)$.

⁸We would only need the exclusion restriction $\lambda_b|(\xi = b, \varepsilon, X, Z) \sim \lambda_b|(\xi = b, Z)$ to hold for realizations of (ε, X) where behavioral model b has multiple solutions.

- (i) *Across behavioral models.*- The predictions for $Pr(Y = y|X)$ produced by any pair of solution configurations across any pair of behavioral models must differ with nonzero probability (with respect to X).
- (ii) *Within behavioral models.*- The predictions for $Pr(Y = y|X)$ for any pair of distinct solution configurations within each behavioral model must differ with nonzero probability.

This would yield a generalization of the full-rank condition we exploited in our previous examples: $\exists y \in \mathcal{Y}$ such that $E[\Xi(y|X, \theta) \cdot \Xi(y|X, \theta)'|Z]$ is invertible for every $\theta \in \Theta$ and a.e Z . Letting $\bar{y}$ be an outcome that satisfies the above restriction and defining $D(y) \equiv \mathbb{1}\{Y = y\}$, the behavioral-solution weights $\delta(Z)$ can be expressed as,

$$\delta(Z) = E[\Xi(\bar{y}|X, \theta_0) \cdot \Xi(\bar{y}|X, \theta_0)'|Z]^{-1} \cdot E[\Xi(\bar{y}|X, \theta_0) \cdot (D(\bar{y}) - g_{\ell_0^{b_0}}^{b_0}(\bar{y}|X, \theta_0))|Z] \text{ a.e } Z.$$

Let

$$\delta(Z, \theta) \equiv E[\Xi(\bar{y}|X, \theta) \cdot \Xi(\bar{y}|X, \theta)'|Z]^{-1} \cdot E[\Xi(\bar{y}|X, \theta) \cdot (D(\bar{y}) - g_{\ell_0^{b_0}}^{b_0}(\bar{y}|X, \theta))|Z],$$

$$\mathbb{P}(y|U, \theta) \equiv p_3^{NC}(y|X, \theta) + \delta(Z, \theta)' \Xi(y|X, \theta)$$

We have $p_Y(y|U) = \mathbb{P}(y|U, \theta_0) \forall y \in \mathcal{Y}$, a.e U , a generalization of the conditional moment restriction in (30). Identifiability of θ_0 can follow if we maintain the assumption that the underlying selection mechanisms assign nonzero weight to at least one behavioral model that is capable of identifying θ_0 . In the examples of our previous sections, this amounted to assuming that ξ selects noncooperative behavior with nonzero probability. A conditional GMM estimator for θ_0 can then be constructed using the approach we described in previous sections. Estimation of the weights $\delta(Z)$, and specification tests can then be pursued along the lines described previously. If we are unwilling to impose restrictions on $\delta(Z)$ that can lead to point-identification of θ_0 , a CS for θ_0 can be based on the conditional moment restriction $p_Y(y|U) = \mathbb{P}(y|U, \theta_0) \forall y \in \mathcal{Y}$, a.e U , which defines θ_0 .

5 A Monte Carlo study

Here we study the performance of the estimator proposed in Section 2.6 in a Monte Carlo experiment. We use three observable scalar payoff covariates, $(W_1, W_2, Z) \stackrel{i.i.d}{\sim} \mathcal{N}(0, 1)$. We parameterize payoffs using $t_1 = \beta_{10}^C + \beta_{10}^W W_1 + \beta_{10}^Z Z$ and $t_2 = \beta_{20}^C + \beta_{20}^W W_2 + \beta_{20}^Z Z$, and constant strategic-interaction effects $(\Delta_{10}, \Delta_{20})$. The unobserved payoff shifters $(\varepsilon_1, \varepsilon_2)$ are bivariate standard Normal with correlation coefficient ρ_0 , independent of (W_1, W_2, Z) . We set $\beta_{10}^C =$

$\beta_{20}^C = 0.5$, $\beta_{10}^W = \beta_{20}^W = -1$, $\beta_{10}^Z = \beta_{20}^Z = 0.5$, and $\rho_0 = 0.5$. The strategic-interaction parameters are $\Delta_{10} = 0.5$ and $\Delta_{20} = 1$. As in Section 2.6, players have two types of possible behavior: cooperative and PSNE. Players cooperate iff $Z + \eta \leq c$, where $\eta \sim \mathcal{N}(0, 1)$ is a “signal” that is independent of all other payoff shifters, and c is a threshold. This satisfies the restrictions in Assumption G2, with $Pr(\text{Cooperation}|Z) \equiv \pi(Z) = E[\mathbb{1}\{\eta \leq c - Z\}|Z] = \Phi(c - Z)$, which is *decreasing* in Z . We set $c = \sqrt{2} \cdot \Phi^{-1}(1/3)$, so the aggregate (i.e, unconditional) probability of cooperation is $\pi \equiv E[\mathbb{1}\{Z + \eta \leq c\}] = \Phi(\sqrt{2} \cdot \Phi^{-1}(1/3)/\sqrt{2}) = 1/3$.

5.1 Estimation results for $\widehat{\theta}$

We applied the estimation procedure in Section 2.6 to 1,000 samples of sizes $n = 500$, $n = 1,500$ and $n = 3,000$. The non-strategic parameters $\gamma \equiv (\beta_1^C, \beta_1^W, \beta_1^Z, \beta_2^C, \beta_2^W, \beta_2^Z, \rho)$ were estimated using the MLE procedure in Section 2.5.1, and the strategic-interaction parameters $\Delta \equiv (\Delta_1, \Delta_2)$ were estimated using the conditional GMM procedure described in Section 2.6, using a Gaussian kernel for Z and a bandwidth of the form $h_n = c_h \cdot \widehat{\sigma}(Z) \cdot n^{-1/5}$. Let $\widehat{f}_Z(z)$ denote the kernel-estimator for $f_Z(z)$, and let $\widehat{f}_{Z,\alpha}$ and $\widehat{\tau}_{Z,\alpha}$ denote the α^{th} sample quantiles of $(\widehat{f}_Z(Z_i))_{i=1}^n$ and $(Z_i)_{i=1}^n$, respectively. Our conditional-GMM inference range was $\mathcal{Z} = \{z \in \mathbb{R} : \widehat{\tau}_{Z,0.005} \leq z \leq \widehat{\tau}_{Z,0.995}, \widehat{f}_Z(z) \geq \widehat{f}_{Z,0.005}\}$. Table 1 summarizes our estimation results using $c_h = 1$ for the conditional-GMM bandwidth, which is close to the so-called “rule of thumb” bandwidth choice. Our estimators perform well, and they estimate both the non-strategic and the strategic parameters with reasonable precision for all sample sizes analyzed. Appendix A4 includes estimation results for $c_h = 0.80$ and $c_h = 1.40$. Even though we do not develop a general theory of bandwidth selection, our estimation results were robust across our bandwidth choices.

Table 1: Monte Carlo results for θ .

Parameter	True value	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
β_1^C	0.5	0.528	0.086	0.511	0.020	0.500	0.010
β_1^W	-1	-1.026	0.106	-1.005	0.018	-1.004	0.008
β_1^Z	0.5	0.515	0.047	0.505	0.010	0.502	0.004
β_2^C	0.5	0.532	0.090	0.502	0.021	0.497	0.010
β_2^W	-1	-1.042	0.150	-1.015	0.019	-0.998	0.009
β_2^Z	0.5	0.519	0.041	0.502	0.009	0.504	0.004
ρ	0.5	0.564	0.108	0.519	0.048	0.499	0.027
Δ_1	0.5	0.610	0.174	0.511	0.084	0.519	0.066
Δ_2	1	1.057	0.225	1.044	0.095	1.033	0.059

• 1,000 simulations, $c_h = 1.0$, Gaussian kernel.

5.2 Estimation results for the probability of cooperation

$\pi(Z)$ is a decreasing function of Z , and we want to analyze if our results capture this feature. Table 2 presents results for $\widehat{\pi}(z, \widehat{\theta})$ for $z = \{-0.862, -0.735, -0.609, -0.483, -0.356\}$ which correspond, approximately, to $\pi(z) = \{0.60, 0.55, 0.50, 0.45, 0.40\}$ respectively. Even though our estimators $\widehat{\pi}(z, \widehat{\theta})$ converge at a nonparametric rate, our results clearly indicate that the probability of cooperation is decreasing in z , and this was true for all sample sizes. Table 3 contains results for the aggregate probability of cooperation $\pi = E_Z[\pi(Z)] = 1/3$, which is reasonably well estimated with $\widehat{\pi} = \frac{1}{n} \sum_{i=1}^n \widehat{\pi}(Z_i, \widehat{\theta})$. Tables 2 and 3 use $c_h = 1$ for our bandwidth. Appendix A4 presents the results for $c_h = 0.80$ and $c_h = 1.40$. Our findings were qualitatively robust across our bandwidth choices.

Table 2: Monte Carlo results for $\widehat{\pi}(z, \widehat{\theta})$ for various values of z .

	True value of $\pi(z)$	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
$z = -0.862$	0.600	0.492	0.083	0.560	0.071	0.563	0.067
$z = -0.735$	0.550	0.466	0.075	0.518	0.067	0.528	0.063
$z = -0.609$	0.500	0.452	0.069	0.478	0.066	0.483	0.061
$z = -0.483$	0.450	0.435	0.063	0.459	0.062	0.449	0.058
$z = -0.356$	0.400	0.412	0.065	0.429	0.061	0.425	0.059

• 1,000 simulations, $c_h = 1.0$, Gaussian kernel.

Table 3: Monte Carlo results for $\widehat{\pi}$

True value of π	$n = 500$		$n = 1,500$		$n = 3,000$	
	median estimate	MSE	median estimate	MSE	median estimate	MSE
$\pi = \frac{1}{3}$	0.341	0.061	0.362	0.055	0.354	0.052

• 1,000 simulations, $c_h = 1.0$, Gaussian kernel.

6 An empirical illustration

A nonparametric test for cooperation in discrete games was proposed in Aradillas-López and Kosenkova (2023). As an illustration, they applied their test to study geographic-

market entry decisions by Lowe’s and Home Depot in the continental United States. Their findings suggested that noncooperative behavior was a better approximation to the outcomes observed in larger markets, while their model of cooperation failed to be rejected by the entry outcomes observed in smaller markets. Their method is not equipped to estimate payoff parameters, nor the probability of cooperation. To have a more complete picture, we revisit this empirical illustration here by applying the approach develop here, assuming cooperation and PSNE as the two candidate behavioral models. We label player $p = 1$ as Lowe’s and $p = 2$ as Home Depot, and we define a market i as a core-based statistical area (CBSA) in the contiguous United States. Our sample consists of $n = 954$ markets.

We label $Y_{pi} = 1$ if player p had presence in market i in 2022 and 0 otherwise. Our payoff covariates include: $X_{pi}^1 \equiv \text{Distance between market } i \text{ and the nearest distribution center of player } p$. $X_i^2 \equiv \text{Small-market indicator}$, equal to one if market i ’s population and number of businesses are both below their medians, and if i is categorized as a micropolitan statistical area. $X_i^3 \equiv \text{Income per household in market } i$. $X_i^4 \equiv \text{Population density in market } i$, and $X_i^5 \equiv \text{Market size}$, measured as $\log(\text{Population}_i)$. Our data comes from Aradillas-López and Kosenkova (2023), and its details are described in Section 6 of that paper. In our parameterization of payoffs, we assume that the slope coefficients of all non-strategic payoff shifters are the same for both players, and we include a player-specific fixed effect in players’ payoffs. Motivated by the findings in Aradillas-López and Kosenkova (2023), we use as our instrument $Z_i = X_i^5 = \text{Market size}$. Unobserved payoff shocks $(\varepsilon_{1i}, \varepsilon_{2i})$ are assumed to be jointly Normal, with mean zero, unit variance, and correlation coefficient ρ_0 . Our specification is consistent with the exclusion restrictions assumed in Section 2.

6.1 Estimation results for $\hat{\theta}$

Estimation results are included in Table 4. The non-strategic parameters are estimated using MLE, as described in Section 2.5.1, and the strategic parameters are estimated using the conditional GMM procedure described in Section 2.6, maintaining the assumption that these firms display noncooperative behavior with nonzero probability. Except for income-per-household, all the non-strategic payoff shifters were statistically significant at a 1% significance level, with the signs we would anticipate for distance-to-distribution-center (-), market size (+), and small-market (-). The estimated correlation coefficient ρ_0 was large (≈ 0.82) and statistically significant, suggesting the presence of market-specific shocks observed by both players subsumed in ε . The tuning parameters (inference range, kernel and bandwidth) were chosen as in our Monte Carlo experiment section, with a bandwidth of the form $h_n = c_h \cdot \hat{\sigma}(Z) \cdot n^{-1/5}$ and $c_h = 1$, yielding $h_n \approx 0.25 \cdot \hat{\sigma}(Z)$. Our results show that the strategic interaction effect estimates were statistically significant, and the estimates suggest that the strategic effect for Lowe’s is larger than that of Home Depot, i.e,

$\Delta_{20} < \Delta_{10}$. To explore this, we tested the null hypothesis $H_0 : \Delta_{20} \geq \Delta_{10}$ and rejected it in favor of $H_1 : \Delta_{20} < \Delta_{10}$ with a p-value of 4.9×10^{-8} . We employed the usual t-statistic based on the asymptotic distribution result in Proposition 1, with the estimator for the asymptotic variance described in Appendix A.

Table 4: Estimation results (standard errors in parentheses)

Non-strategic parameters	
Player 1 (Lowe's) fixed effect	Player 2 (Home Depot) fixed effect
3.226* (0.552)	2.767* (0.804)
Distance to nearest distribution center	Small-market indicator
-0.323* (0.054)	-0.695* (0.198)
Income per household	Payroll per business establishment
-0.100 (0.088)	-0.231* (0.043)
Population density	Market size
-0.443* (0.174)	0.467* (0.146)
ρ_0	
0.824* (0.263)	
Strategic-interaction parameters	
Player 1 (Lowe's): Δ_{10}	Player 2 (Home Depot): Δ_{20}
0.971* (0.091)	0.278* (0.022)

(*) denotes statistically significant at a 1% significance level.

6.2 Probability of cooperation and market size

The nonparametric test in Aradillas-López and Kosenkova (2023) suggested evidence that noncooperative behavior is a better approximation to the observed outcomes in larger markets relative to smaller ones. In our context, this conjecture would lead us to expect that $\pi(Z)$ (the probability of cooperation) is decreasing in Z (market size). Let $Z_{(\tau)}$ denote the τ^{th} quantile of Z (market size). In Table 5 we present estimates for $E[\pi(Z)|Z > Z_{(\tau)}]$, which we estimate with $\widehat{E}[\widehat{\pi}(Z, \widehat{\theta})|Z > Z_{(\tau)}]$. Our results suggest that this quantity is decreasing in $Z_{(\tau)}$, providing support to the conjecture that the outcomes observed in larger market

are more likely to have been produced by noncooperative behavior, while our cooperation model approximates the outcomes in smaller markets with higher likelihood.

Table 5: Estimates for $E[\pi(Z)|Z > Z_{(\tau)}]$

	$Z_{(0.33)}$	$Z_{(0.50)}$	$Z_{(0.66)}$	$Z_{(0.75)}$	$Z_{(0.85)}$	$Z_{(0.90)}$
$\widehat{E}[\widehat{\pi}(Z, \widehat{\theta}) Z > Z_{(\tau)}]$	0.283	0.199	0.121	0.088	0.053	0.038

7 Concluding remarks

While allowing for multiple equilibria, the econometric analysis of games with non experimental data typically maintains the assumption that every observation was produced by the same behavioral model. This paper aims to relax this restriction by assuming instead a collection of candidate behavioral models, each one with potentially multiple solutions, that could have produced each observation. In static discrete games with parametric normal-form representations, we show that if there is an observable instrument Z that controls for the dependence between the underlying selection mechanisms and the players' payoff functions, the predictions of our global model can be written as a convolution of the predictions of the candidate solutions, with the convolution weights being nonparametric functionals of Z . If there exists an outcome for which our candidate solutions are not observationally equivalent, these behavioral weights can be estimated in a first step, and the behavioral convolution can be used for estimation and inference of the game's normal-form parameters. Point identification of these parameters can result if the behavioral selection mechanism assigns nonzero probability to behavioral models that have sufficient identification power. Because the predictions of our model boil down to conditional moment restrictions, the validity of any candidate instrument Z can be tested through existing semiparametric specification tests. A conditional GMM estimator was proposed, and Monte Carlo experiments showed that the resulting estimates for normal-form parameters and behavioral weights had good properties. As an empirical illustration we analyzed geographic entry decisions by Lowe's and Home Depot. Assuming cooperation and pure-strategy Nash equilibrium as the two candidate behavioral models, we found evidence of statistically significant strategic-interaction effects, and we also found evidence consistent with the presence of both behavioral models across the data, with the probability of cooperation decreasing with market size.

Appendix A

A1 Deriving the expression for $\frac{\partial^2 Q_S(\theta)}{\partial \Delta \partial \Delta'}$ in Section 2.5.2

From the definition of $\partial \mu_S(U, \theta)$, we have

$$\frac{\partial \mu_S(U, \theta)}{\partial \Delta_p} = \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \Delta_p} + \pi(Z, \theta) \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \Delta_p} + \frac{\partial \pi(Z, \theta)}{\partial \Delta_p} \cdot \Xi_S(X, \theta). \quad (\text{A.1})$$

And grouping $\frac{\partial \mu_S(U, \theta)}{\partial \Delta} = \left(\frac{\partial \mu_S(U, \theta)}{\partial \Delta_1}, \frac{\partial \mu_S(U, \theta)}{\partial \Delta_2} \right)'$,

$$\begin{aligned} \frac{\partial^2 Q_S(\theta)}{\partial \Delta \partial \Delta'} &= E \left[\frac{\partial^2 \tau(U, \theta)}{\partial \Delta \partial \Delta'} \cdot \tau(U, \theta) \right] + E \left[\frac{\partial \tau(U, \theta)}{\partial \Delta} \cdot \frac{\partial \tau(U, \theta)}{\partial \Delta} \right]' \\ &= - \int E \left[\frac{\partial^2 \mu_S(U, \theta)}{\partial \Delta \partial \Delta'} \cdot \mathbb{1}\{U \leq u\} \right] \cdot E[\varphi_S(V, \theta) \cdot \mathbb{1}\{U \leq u\}] dF_U(u) \\ &\quad + \int E \left[\frac{\partial \mu_S(U, \theta)}{\partial \Delta} \cdot \mathbb{1}\{U \leq u\} \right] \cdot E \left[\frac{\partial \mu_S(U, \theta)}{\partial \Delta} \cdot \mathbb{1}\{U \leq u\} \right]' dF_U(u). \end{aligned}$$

At $\theta = \theta_0$, this simplifies to

$$\frac{\partial^2 Q_S(\theta_0)}{\partial \Delta \partial \Delta'} = \int E \left[\frac{\partial \mu_S(U, \theta_0)}{\partial \Delta} \cdot \mathbb{1}\{U \leq u\} \right] \cdot E \left[\frac{\partial \mu_S(U, \theta_0)}{\partial \Delta} \cdot \mathbb{1}\{U \leq u\} \right]' dF_U(u).$$

Invertibility of $\frac{\partial^2 Q_S(\theta_0)}{\partial \Delta \partial \Delta'}$ requires $\int E \left[\frac{\partial \mu_S(U, \theta)}{\partial \Delta} \cdot \mathbb{1}\{U \leq u\} \right] \cdot E \left[\frac{\partial \mu_S(U, \theta)}{\partial \Delta} \cdot \mathbb{1}\{U \leq u\} \right]' dF_U(u)$ to have full rank in an open neighborhood of θ_0 . Basic conditions for invertibility can be derived by simplifying the expression for $\frac{\partial^2 Q_S(\theta_0)}{\partial \Delta \partial \Delta'}$. We proceed by analyzing each term in (A.1). For $p = 1, 2$, denote $\nabla_p F_{1,2}(\epsilon_1, \epsilon_2 | \rho) \equiv \frac{\partial F_{1,2}(\epsilon_1, \epsilon_2 | \rho)}{\partial \epsilon_p}$ and let

$$\begin{aligned} H_p^{NC}(X, \theta) &\equiv \nabla_p F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1, X_2^{ns'} \beta_2 - X_2^{s'} \Delta_2 | \rho), \\ H^C(X, \theta) &\equiv \sum_{q=1}^2 \nabla_q F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2, X_2^{ns'} \beta_2 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2 | \rho). \end{aligned}$$

From (2),

$$\begin{aligned} \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \Delta_p} &= -H_p^{NC}(X, \theta) \cdot X_p^s, \quad \frac{\partial \mu_S^C(X, \theta)}{\partial \Delta_p} = -H^C(X, \theta) \cdot X_p^s, \\ \frac{\partial \Xi_S(X, \theta)}{\partial \Delta_p} &= (H_p^{NC}(X, \theta) - H^C(X, \theta)) \cdot X_p^s \end{aligned} \quad (\text{A.2})$$

From here, (9) yields,

$$\begin{aligned} \frac{\partial \pi(Z, \theta)}{\partial \Delta_p} &= \frac{E \left[-\frac{\partial \mu_S^{NC}(X, \theta)}{\partial \Delta_p} \cdot \Xi_S(X, \theta) + \left(S - \mu_S^{NC}(X, \theta) \right) \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \Delta_p} \mid Z \right]}{E \left[\Xi_S(X, \theta)^2 \mid Z \right]} \\ &\quad - 2 \cdot \frac{E \left[\left(S - \mu_S^{NC}(X, \theta) \right) \cdot \Xi_S(X, \theta) \mid Z \right] \cdot E \left[\Xi_S(X, \theta) \frac{\partial \Xi_S(X, \theta)}{\partial \Delta_p} \mid Z \right]}{\left(E \left[\Xi_S(X, \theta)^2 \mid Z \right] \right)^2} \end{aligned} \quad (\text{A.3})$$

From the representation in (6), evaluated at $\theta = \theta_0$ we obtain

$$\begin{aligned} E \left[\left(S - \mu_S^{NC}(X, \theta) \right) \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \Delta_p} \mid Z \right] &= \pi(Z) \cdot E \left[\Xi_S(X, \theta_0) \cdot \frac{\partial \Xi_S(X, \theta_0)}{\partial \Delta_p} \mid Z \right], \\ E \left[\left(S - \mu_S^{NC}(X, \theta_0) \right) \cdot \Xi_S(X, \theta_0) \mid Z \right] &= \pi(Z) \cdot E \left[\Xi_S(X, \theta_0)^2 \mid Z \right] \end{aligned} \quad (\text{A.4})$$

Denote $J_p(U, \theta) \equiv \pi(Z, \theta) \cdot H^C(X, \theta) + (1 - \pi(Z, \theta)) \cdot H_p^{NC}(X, \theta)$. Plugging (A.2) and (A.4) into (A.3) and grouping terms, we obtain

$$\frac{\partial \pi(Z, \theta_0)}{\partial \Delta_p} = \frac{E \left[J_p(U, \theta_0) \cdot \Xi_S(X, \theta_0) \cdot X_p^S \mid Z \right]}{E \left[\Xi_S(X, \theta_0)^2 \mid Z \right]}$$

From here, (A.1) yields

$$\frac{\partial \mu_S(U, \theta_0)}{\partial \Delta_p} = -J_p(U, \theta_0) \cdot X_p^S + \frac{E \left[J_p(U, \theta_0) \cdot \Xi_S(X, \theta_0) \cdot X_p^S \mid Z \right]}{E \left[\Xi_S(X, \theta_0)^2 \mid Z \right]} \cdot \Xi_S(X, \theta_0)$$

Thus, letting

$$\dot{\mu}_{S, \Delta}(U, \theta) \equiv \begin{pmatrix} -J_1(U, \theta) \cdot X_1^S + \frac{E \left[J_1(U, \theta) \cdot \Xi_S(X, \theta) \cdot X_1^S \mid Z \right]}{E \left[\Xi_S(X, \theta)^2 \mid Z \right]} \cdot \Xi_S(X, \theta) \\ -J_2(U, \theta) \cdot X_2^S + \frac{E \left[J_2(U, \theta) \cdot \Xi_S(X, \theta) \cdot X_2^S \mid Z \right]}{E \left[\Xi_S(X, \theta)^2 \mid Z \right]} \cdot \Xi_S(X, \theta) \end{pmatrix}$$

Then, $\frac{\partial^2 Q_S(\theta_0)}{\partial \Delta \partial \Delta'} = \int E \left[\dot{\mu}_{S, \Delta}(U, \theta_0) \cdot \mathbb{1}\{U \leq u\} \right] \cdot E \left[\dot{\mu}_{S, \Delta}(U, \theta_0)' \cdot \mathbb{1}\{U \leq u\} \right] dF_U(u)$. This is the expression in equation (12). ■

A2 Proof of Result M

Recall that $\Xi(y|X, \theta) \equiv (m_1(y|X, \theta), m_2(y|X, \theta), m_3(y|X, \theta))'$ where,

$$\begin{aligned} m_1(y|X, \theta) &\equiv p^C(y|X, \theta) - p_3^{NC}(y|X, \theta) = p^C(y|X, \theta) - (g_{\mathcal{U}}^{NE}(y|X, \theta) + g_{\mathcal{M},3}^{NE}(y|X, \theta)), \\ m_2(y|X, \theta) &\equiv p_3^{NC}(y|X, \theta) - p_1^{NC}(y|X, \theta) = g_{\mathcal{M},3}^{NE}(y|X, \theta) - g_{\mathcal{M},1}^{NE}(y|X, \theta), \\ m_3(y|X, \theta) &\equiv p_3^{NC}(y|X, \theta) - p_2^{NC}(y|X, \theta) = g_{\mathcal{M},3}^{NE}(y|X, \theta) - g_{\mathcal{M},2}^{NE}(y|X, \theta). \end{aligned}$$

And,

$$\begin{aligned} p^C(y|X, \theta_0) &\equiv \int \mathbb{1}\{\varepsilon \in \mathcal{R}^C(y|X, \beta, \Delta)\} f_{1,2}(\varepsilon|\rho) d\varepsilon, \\ g_{\mathcal{U}}^{NE}(y|X, \theta) &\equiv \int \mathbb{1}\{\varepsilon \in \mathcal{U}^{NE}(y|X, \beta, \Delta)\} f_{1,2}(\varepsilon|\rho) d\varepsilon, \\ g_{\mathcal{M},\ell}^{NE}(y|X, \theta) &\equiv \int \sigma_{\mathcal{M},\ell}^{NE}(y|X, \varepsilon, \beta, \Delta) \cdot \mathbb{1}\{\varepsilon \in \mathcal{M}^{NE}(X, \beta, \Delta)\} f_{1,2}(\varepsilon|\rho) d\varepsilon. \end{aligned}$$

Recall from (23) that $\sigma_{\mathcal{M},1}^{NE}(y|X, \varepsilon, \theta) = \mathbb{1}\{y = (1, 0)\}$, $\sigma_{\mathcal{M},2}^{NE}(y|X, \varepsilon, \theta) = \mathbb{1}\{y = (0, 1)\}$, and

$$\sigma_{\mathcal{M},3}^{NE}(y|X, \varepsilon, \theta) = \left(\frac{X_2^{ns'} \beta_2 - \varepsilon_2}{X_2^{s'} \Delta_2} \right)^{y_1} \left(1 - \frac{X_2^{ns'} \beta_2 - \varepsilon_2}{X_2^{s'} \Delta_2} \right)^{1-y_1} \left(\frac{X_1^{ns'} \beta_1 - \varepsilon_1}{X_1^{s'} \Delta_1} \right)^{y_2} \left(1 - \frac{X_1^{ns'} \beta_1 - \varepsilon_1}{X_1^{s'} \Delta_1} \right)^{1-y_2}$$

Let $G_{\mathcal{M}}(X, \theta) = \int \mathbb{1}\{\varepsilon \in \mathcal{M}^{NE}(X, \beta, \Delta)\} f_{1,2}(\varepsilon|\rho) d\varepsilon$ (the conditional probability of falling in the multiple-NE region when $\theta_0 = \theta$). Note that $g_{\mathcal{M},1}^{NE}(y|X, \theta) = \mathbb{1}\{y = (1, 0)\} \cdot G_{\mathcal{M}}(X, \theta)$ and $g_{\mathcal{M},2}^{NE}(y|X, \theta) = \mathbb{1}\{y = (0, 1)\} \cdot G_{\mathcal{M}}(X, \theta)$. Thus, $g_{\mathcal{M},1}^{NE}(y|X, \theta) = g_{\mathcal{M},2}^{NE}(y|X, \theta) = 0$ for $y \in \{(0, 0), (1, 1)\}$ and therefore,

$$\Xi(y|X, \theta) = \begin{pmatrix} p^C(y|X, \theta) - (g_{\mathcal{U}}^{NE}(y|X, \theta) + g_{\mathcal{M},3}^{NE}(y|X, \theta)) \\ g_{\mathcal{M},3}^{NE}(y|X, \theta) \\ g_{\mathcal{M},3}^{NE}(y|X, \theta) \end{pmatrix} \text{ for } y \in \{(0, 0), (1, 1)\}.$$

Thus, $\Xi(y|X, \theta) \cdot \Xi(y|X, \theta)'$ can never have full rank for $y = (0, 0)$ or $y = (1, 1)$, and Result M cannot hold for these outcomes. Now, for $y = (1, 0)$,

$$\Xi((1, 0)|X, \theta) = \begin{pmatrix} p^C((1, 0)|X, \theta) - (g_{\mathcal{U}}^{NE}((1, 0)|X, \theta) + g_{\mathcal{M},3}^{NE}((1, 0)|X, \theta)) \\ g_{\mathcal{M},3}^{NE}((1, 0)|X, \theta) - G_{\mathcal{M}}(X, \theta) \\ g_{\mathcal{M},3}^{NE}((1, 0)|X, \theta) \end{pmatrix}$$

and for $y = (0, 1)$,

$$\Xi((0, 1)|X, \theta) = \begin{pmatrix} p^C((0, 1)|X, \theta) - (g_{\mathcal{U}}^{NE}((0, 1)|X, \theta) + g_{\mathcal{M}, 3}^{NE}((0, 1)|X, \theta)) \\ g_{\mathcal{M}, 3}^{NE}((0, 1)|X, \theta) \\ g_{\mathcal{M}, 3}^{NE}((0, 1)|X, \theta) - G_{\mathcal{M}}(X, \theta) \end{pmatrix},$$

The conditions in Assumption G3 and the restrictions to the parameter space Θ in (4) imply that, for each $y \in \mathcal{Y}$, $\theta \in \Theta$, and a.e Z , the support of

$$(p^C(y|X, \theta), G_{\mathcal{M}}(X, \theta), g_{\mathcal{U}}^{NE}(y|X, \theta), g_{\mathcal{M}, 3}^{NE}(y|X, \theta))',$$

conditional on Z , is not contained in any proper linear subspace of $\mathbb{R}^4$. Since this is true in particular for $y \in \{(1, 0), (0, 1)\}$, it follows that $E[\Xi(y|X, \theta) \cdot \Xi(y|X, \theta)'|Z]$ has full rank and is invertible for every $\theta \in \Theta$ and a.e Z for $y \in \{(1, 0), (0, 1)\}$. ■

A3 Estimating Ω_{Δ} , the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$

Using Proposition 1, our proposal is to estimate Ω_{Δ} , the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$, as $\widehat{\Omega}_{\Delta} \equiv \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta})'$, where $\widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta})$ is an estimator of the influence function $\psi_{\Delta, n}(V_i; \theta_0)$, described in (22). Let $\widehat{R}_a^{\pi}(z, \theta) \equiv \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right)$ and $\widehat{R}_b^{\pi}(z, \theta) \equiv \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right)$ and note that $\widehat{\pi}(z, \theta) = \frac{\widehat{R}_a^{\pi}(z, \theta)}{\widehat{R}_b^{\pi}(z, \theta)}$. We estimate the influence function $\psi_n^{\pi}(V_i; z, \theta)$ in (19) as,

$$\begin{aligned} \widehat{\psi}_n^{\pi}(V_i; z, \theta) &\equiv \frac{1}{\widehat{R}_b^{\pi}(z, \theta)} \cdot \widehat{\psi}_{a, n}^{\pi}(V_i; z, \theta) - \frac{\widehat{\pi}(z, \theta)}{\widehat{R}_b^{\pi}(z, \theta)} \cdot \widehat{\psi}_{b, n}^{\pi}(U_i; z, \theta), \quad \text{where} \\ \widehat{\psi}_{a, n}^{\pi}(V_i; z, \theta) &\equiv (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - h_n^{d_Z} \cdot \widehat{R}_a^{\pi}(z, \theta), \\ \widehat{\psi}_{b, n}^{\pi}(U_i; z, \theta) &\equiv \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right) - h_n^{d_Z} \cdot \widehat{R}_b^{\pi}(z, \theta) \end{aligned}$$

Next, we estimate⁹ $\frac{\partial^2 \widehat{Q}_{S,Z}(\theta)}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{j=1}^n \left(\frac{\partial^2 \widehat{\tau}_Z(U_j, \theta)}{\partial \theta \partial \theta'} \cdot \widehat{\tau}_Z(U_j, \theta) + \frac{\partial \widehat{\tau}_Z(U_j, \theta)}{\partial \theta} \cdot \frac{\partial \widehat{\tau}_Z(U_j, \theta)}{\partial \theta} \right)$, where, $\frac{\partial \widehat{\tau}_Z(u, \theta)}{\partial \theta} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{U_i \leq u\}$, $\frac{\partial^2 \widehat{\tau}_Z(u, \theta)}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \widehat{\varphi}_S(V_i, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{U_i \leq u\}$ and,

$$\begin{aligned} \widehat{\varphi}_S(V_i, \theta) &\equiv S_i - \mu_S^{NC}(X_i, \theta) - \widehat{\pi}(Z_i, \theta) \cdot \Xi_S(X_i, \theta), \\ \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \theta} &= - \left(\frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta} + \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \cdot \Xi_S(X_i, \theta) + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \cdot \widehat{\pi}(Z_i, \theta) \right), \\ \frac{\partial^2 \widehat{\varphi}_S(V_i, \theta)}{\partial \theta \partial \theta'} &= - \left(\frac{\partial^2 \mu_S^{NC}(X_i, \theta)}{\partial \theta \partial \theta'} + \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} + \frac{\partial^2 \widehat{\pi}(Z_i, \theta)}{\partial \theta \partial \theta'} \cdot \Xi_S(X_i, \theta) \right. \\ &\quad \left. + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \cdot \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} + \frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta \partial \theta'} \cdot \widehat{\pi}(Z_i, \theta) \right). \end{aligned}$$

The Jacobian $\frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta}$ and Hessian $\frac{\partial^2 \widehat{\pi}(Z_i, \theta)}{\partial \theta \partial \theta'}$ of $\widehat{\pi}(Z_i, \theta)$ are estimated as,

$$\begin{aligned} \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} &= \frac{1}{\widehat{R}_b^\pi(z, \theta)} \cdot \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} \right), \\ \frac{\partial^2 \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= - \frac{1}{\widehat{R}_b^\pi(z, \theta)^2} \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j} \cdot \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \\ &\quad + \frac{1}{\widehat{R}_b^\pi(z, \theta)} \cdot \left(\frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_j} \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right), \end{aligned}$$

⁹Note that $\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \theta \partial \theta'} = E \left[\frac{\partial \tau_Z(U, \theta_0)}{\partial \theta} \cdot \frac{\partial \tau_Z(U, \theta_0)}{\partial \theta} \right]$ (since $\tau_Z(U, \theta_0) = 0$ a.s), and we can estimate $\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \theta \partial \theta'}$ as

$$\frac{\partial^2 \widehat{Q}_{S,Z}(\widehat{\theta})}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{j=1}^n \frac{\partial \widehat{\tau}_Z(U_j, \widehat{\theta})}{\partial \theta} \cdot \frac{\partial \widehat{\tau}_Z(U_j, \widehat{\theta})}{\partial \theta}.$$

where,

$$\begin{aligned}
\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} &= \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) - \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \right) \cdot K\left(\frac{Z_i - z}{h_n}\right) \\
\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} &= \frac{2}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right), \\
\frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) - \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_j} \right. \\
&\quad \left. - \frac{\partial^2 \mu_S^{NC}(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) - \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \right) \cdot K\left(\frac{Z_i - z}{h_n}\right), \\
\frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= \frac{2}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \right) \cdot K\left(\frac{Z_i - z}{h_n}\right).
\end{aligned}$$

For $m \in \mathbb{N}$ let $(m)_k \equiv m \cdot (m-1) \cdots (m-k)$. From the definition in (21), we estimate

$$\begin{aligned}
\widehat{\Gamma}_{\Delta, n}(V_i; \theta) &= \widehat{\varphi}_S(V_i, \theta) \cdot \frac{1}{(n-1)_1} \sum_{j \neq i} \sum_{k \neq i, j} \frac{\partial \widehat{\varphi}_S(V_k, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_k \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \\
&\quad - \frac{1}{(n-1)_2} \sum_{j \neq i} \sum_{k \neq i, j} \sum_{\ell \neq i, j, k} \frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} \cdot \Xi_S(X_j, \theta) \cdot \frac{1}{h_n^{d_Z}} \widehat{\psi}_n^\pi(V_i; Z_j, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_j \leq U_k\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_k\}
\end{aligned}$$

Let $\widehat{\psi}_\gamma(V_i; \widehat{\gamma})$ be the estimated MLE influence function for $\widehat{\gamma}$. Using (22), we estimate¹⁰,

$$\widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \equiv - \frac{\partial^2 \widehat{Q}_{S, \mathcal{Z}}(\widehat{\theta})}{\partial \Delta \partial \Delta'}^{-1} \times \left(\widehat{\Gamma}_{\Delta, n}(V_i; \widehat{\theta}) + \frac{\partial^2 \widehat{Q}_{S, \mathcal{Z}}(\widehat{\theta})}{\partial \Delta \partial \gamma'} \cdot \widehat{\psi}_\gamma(V_i; \widehat{\gamma}) \right)$$

From here, we estimate Ω_Δ , the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$, as $\widehat{\Omega}_\Delta \equiv \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta})'$. In Section S1 of the Econometric Supplement we show that, under the restrictions of Proposition 1,

$$\left\| \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta})' - \frac{1}{n} \sum_{i=1}^n \psi_{\Delta, n}(V_i; \theta_0) \cdot \psi_{\Delta, n}(V_i; \theta_0)' \right\| \xrightarrow{p} 0,$$

so $\|\widehat{\Omega}_\Delta - \Omega_\Delta\| \xrightarrow{p} 0$. Once again, these results are obtained using standard arguments. ■

¹⁰See footnote 9.

A4 Additional Monte Carlo results

Next, we compare our Monte Carlo experiment results across alternative bandwidth choices. Specifically, for the bandwidths produced by $c_h = 0.80$, $c_h = 1.0$ (the results shown in the paper), and for $c_h = 1.40$. Our MLE estimates for the non-strategic parameters of the model do not depend on the bandwidth choice, so the results for these parameters remain as described in Table 1 regardless of bandwidth selection. Table A1 presents the estimates of the strategic interaction parameters (Δ_1, Δ_2) for each bandwidth choice. The results are qualitatively very similar, across all bandwidths, particularly for $n = 3,000$, but the results are also relatively close for our smaller sample sizes.

Table A1: Monte Carlo estimation results for (Δ_1, Δ_2) for $c_h \in \{0.80, 1.0, 1.40\}$

Results for $c_h = 0.80$							
Parameter	True value	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
Δ_1	0.5	0.608	0.179	0.508	0.084	0.512	0.064
Δ_2	1	1.054	0.230	1.042	0.092	1.029	0.058
Results for $c_h = 1.00$							
Parameter	True value	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
Δ_1	0.5	0.610	0.174	0.511	0.084	0.519	0.066
Δ_2	1	1.057	0.225	1.044	0.095	1.033	0.059
Results for $c_h = 1.40$							
Parameter	True value	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
Δ_1	0.5	0.619	0.182	0.521	0.086	0.512	0.067
Δ_2	1	1.044	0.239	1.051	0.103	1.027	0.067

• 1,000 simulations in each case, Gaussian kernel.

Table A2 presents results for the estimated behavioral weights $\pi(Z)$ for the same collection of values of Z studied in Table 2 of our Monte Carlo section. For all bandwidth choices, our estimates clearly show that $\pi(Z)$ is decreasing in Z , which was a key feature of our designs. Each bandwidth selected did a reasonable job of estimating $\pi(Z)$, with oversmoothing yielding slightly better results in terms of MSE, although all bandwidth choices were comparable in their performance.

Table A2: Monte Carlo results for $\widehat{\pi}(z, \widehat{\theta})$ for various values of z .

Results for $c_h = 0.80$							
	True value of $\pi(z)$	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
$z = -0.862$	0.600	0.513	0.079	0.553	0.071	0.564	0.072
$z = -0.735$	0.550	0.485	0.075	0.532	0.066	0.521	0.063
$z = -0.609$	0.500	0.474	0.074	0.479	0.063	0.489	0.063
$z = -0.483$	0.450	0.456	0.068	0.448	0.063	0.444	0.059
$z = -0.356$	0.400	0.410	0.067	0.412	0.063	0.424	0.061
Results for $c_h = 1.00$							
	True value of $\pi(z)$	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
$z = -0.862$	0.600	0.492	0.083	0.560	0.071	0.563	0.067
$z = -0.735$	0.550	0.466	0.075	0.518	0.067	0.528	0.063
$z = -0.609$	0.500	0.452	0.069	0.478	0.066	0.483	0.061
$z = -0.483$	0.450	0.435	0.063	0.459	0.062	0.449	0.058
$z = -0.356$	0.400	0.412	0.065	0.429	0.061	0.425	0.059
Results for $c_h = 1.40$							
	True value of $\pi(z)$	$n = 500$		$n = 1,500$		$n = 3,000$	
		median estimate	MSE	median estimate	MSE	median estimate	MSE
$z = -0.862$	0.600	0.428	0.092	0.537	0.068	0.535	0.066
$z = -0.735$	0.550	0.418	0.080	0.495	0.063	0.501	0.062
$z = -0.609$	0.500	0.410	0.071	0.462	0.059	0.466	0.059
$z = -0.483$	0.450	0.381	0.068	0.421	0.059	0.434	0.056
$z = -0.356$	0.400	0.357	0.062	0.383	0.058	0.401	0.054

• 1,000 simulations in each case, Gaussian kernel.

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